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THE GENERATION AND MODULATION OF ENDOGENOUS RHYTHMICITY IN THE *APLYSIA* BURSTING PACEMAKER NEURONE R15

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I. INTRODUCTION

Rhythmic activity in animals has always attracted the attention of biologists. This is in part because it is so readily recognized and so widespread. Furthermore, for the purposes of analysis, repetitive behaviour has advantages over other forms of animal function. It can be fitted to the well-developed formalisms of physical oscillator theory and regularly-recurring, similarly-structured phenomena are particularly suited to reveal the effects of phasic or tonic experimental perturbations (Moffett, 1977). It is therefore not surprising that rhythmic action potential production was soon noticed by electrophysiologists.

In the early 1940s, Arvanitaki recorded electrical activity extracellularly from the large and readily accessible neuronal somata in the abdominal ganglion of *Aplysia*, an opisthobranch mollusc (Arvanitaki and Cardot, 1941a,b). A few years later, Tauc (1955), and Arvanitaki and

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Chalazonitis (1955a,b, 1958) utilized the then-novel method of intracellular recording via a glass micropipette to monitor the activity of *Aplysia* neurones, including an apparently autoactive "giant" cell which produced bursts of action potentials at regular intervals of a few seconds. Many of the neuronal somata in the *Aplysia* ganglia have now been mapped and labelled (Coggeshall *et al.*, 1966; Frazier *et al.*, 1967), and the most commonly studied endogenous burster is known as R15. Indeed, next to the squid giant axon, R15 is probably the most intensively studied identified neurone.

Our intention in this paper is to review the considerable amount of information available on the structure, function and properties of R15, and, on the basis of its measured characteristics, to present a new and comprehensive model for the ionic basis of rhythmic bursting activity and its modulation in R15. Specificity will be a major theme of this paper, since it is now clear that superficially similar neurones in the molluscs respond in very different ways to the same experimental stimuli. We have therefore drawn our data as far as possible from experiments carried out on R15 itself, although from time to time we compare observations made on other molluscan pacemaker neurones. It follows that the solution we offer to the problem of the ionic basis of bursting refers specifically to R15 although we hope it will have some relevance to understanding the endogenous activity of molluscan neurones in general.

II. NEURONE R15

1. *Identification, Location and Morphology*

In the *Aplysia* abdominal (parietovisceral) ganglion, the neuronal cell bodies are near the surface of the ganglion with the synaptic region located in a central neuropil. The remarkably large size of many of the cell bodies early attracted attention (e.g. Mazzarelli, 1893) and Arvanitaki suggested the possibility of identifying individual cells on the basis of position (Arvanitaki and Cardot, 1941a; Arvanitaki and Tchou, 1942) and later of firing pattern as well (Arvanitaki and Chalazonitis, 1958).

Frazier *et al.* (1967) systematically identified the larger neurones of the abdominal ganglion on the basis of position, firing pattern, size and colour, as well as by their responses to synaptic input and iontophoretic application of acetylcholine. They utilized *Aplysia californica*, the species most often studied for neurophysiological purposes, but there appears to be little difference in the morphology, spontaneous activity, and synaptic responses among the abdominal ganglion neurones of different species of *Aplysia* (Blankenship and Coggeshall, 1976). The designation R15 has been given to a caudal cell, always seen on the dorsal surface, which is usually the largest cell in the right caudal quarter of the abdominal ganglion. Its soma diameter varies from 300 to 500 μm in large animals (Winlow and Kandel, 1976; our observations) and it is usually whitish in large animals but more orange coloured in smaller animals because of the smaller number of granules in its cytoplasm (Frazier *et al.*, 1967). On rare occasions R15 has been found duplicated or even triplicated (Dagan and Adams, 1981). R15 has been referred to as the Br cell (Arvanitaki and Chalazonitis, 1958, 1961a,b), the parabolic burster (Strumwasser, 1965) and Oberon (L. Tauc and colleagues).

The morphology of R15 is rather variable (Frazier *et al.*, 1967; Winlow and Kandel, 1976). It sends axonal branches along the pericardial nerve and usually along the siphon, genital and spermathecal nerves as well as the left connective of the abdominal ganglion. The axon is branched in a manner that varies from animal to animal. The dendritic tree arises from the soma or axon hillock (Winlow and Kandel, 1976). The various axonal branches appear to have different conduction velocities since cuff electrode recordings made from the genital and pericardial nerves of intact, freely-moving animals (Parsons *et al.*, 1983) reveal fractionation of the axonal compound action potentials recorded later in the burst (Maat *et al.*, 1983). It is possible that the locus of pure Na^+ spike initiation, which is axonal in most neurons (Treistman, 1978, 1980), may be altered by repetitive activity (Mooney and Waziri, 1984). In Section III we will show that axonal action potential conduction and the associated current flow is an essential part of somal burst generation in R15.

Infolding of the somal membrane is another morphological feature which contributes to

the form of R15's bursting activity. In common with other *Aplysia* neurones (Batham, 1961; Bullock, 1961; Rosenbluth, 1963; Coggeshall, 1967), R15 probably has extensive membrane indentations which could contribute to the accumulation of K^+ ions close to the soma during bursting activity (Eaton, 1972; Ascher, 1972; Lux and Heyer, 1975; Gorman *et al.*, 1982a). Indeed, it has been suggested that endogenous bursters show a greater degree of somal membrane infolding than non-bursters (Coggeshall, 1967). In Section III we give details of the probable shift in K^+ equilibrium potential due to extracellular K^+ accumulation, and the effect of this shift on bursting activity.

Synaptic input onto R15 occurs in the central neuropil of the abdominal ganglion and probably other ganglia, but not directly onto the soma (Rosenbluth, 1963; Coggeshall, 1967). Axosomatic synapses apparently never occur in adult *Aplysia* although they have been detected at early developmental stages (Schacher *et al.*, 1979). However, some somata in the abdominal ganglion are associated with fibres which might bear release sites for neuroactive substances (Kistler *et al.*, 1983; Ono and McCaman, 1984). As we will describe in detail in Section IV, the somal membrane of R15 possesses receptors for several neurotransmitters, and responses to neurotransmitters presumably acting via receptors in the neuropil can have a profound influence on currents mediated by somal ion channels. The origins and properties of the synaptic inputs onto R15 will be detailed in Section IV.

2. *Function*

Although we are not primarily concerned here with the function of R15 in the nervous system of *Aplysia*, it seems worthwhile to summarize what is known on the subject. In particular, it is possible to make some sense of the array of inputs to which R15 responds with changes in its bursting activity. Virtually nothing is known of the destination of R15's output.

R15 is often referred to as a neurosecretory cell and bursting activity does seem to be a characteristic of some neurosecretory systems in the molluscs and other groups (Berlind, 1977; Pinsker and Dudek, 1977; Cooke, 1977, 1981). The evidence is suggestive but still incomplete. Scharrer (1935a,b) was the first to report the existence of nerve cells exhibiting some neurosecretory properties in the abdominal ganglion of *Aplysia*. Rosenbluth (1963) and Simpson *et al.* (1963) detected granules of an apparently neurosecretory nature in abdominal neurones, and this finding was confirmed by Coggeshall *et al.* (1966; Coggeshall, 1967) and Frazier *et al.* (1967). In R15, the granules are less electron dense than in the other cells and are 150 –250 nm in diameter. They are located in the soma, distant from the nerve terminals. This is a characteristic of neurosecretory neurones and is presumed to reflect the storage of secretory material prior to its processing and transport to the terminals. Unlike other neurosecretory cells in the abdominal ganglion, R15 has no terminals in the ganglionic sheath (Frazier *et al.*, 1967).

A considerable amount of analysis has been carried out on the identified neurones of *Aplysia*, including R15, in a search for distinct patterns of protein concentration and turnover that might characterize particular neurones and correlate with bursting, neurosecretion and other properties. Initial studies by Wilson (1971) and Gainer (1971) showed that certain neurones, R15 among them, can devote as much as 30% of their leucine incorporation to the synthesis of one or a few peptides of approximately 12 K daltons and 6–9 K daltons. These findings have been refined and confirmed (Wilson, 1974; Gainer *et al.*, 1975; Gainer and Barker, 1975). The production of these peptides appears to be correlated with a possible neurosecretory function (Wilson, 1971, 1976; Gainer and Wollberg, 1974; Wilson and Strumwasser, 1975) and not with bursting *per se* (Berry, 1975) as originally suggested by Strumwasser (1965, 1967a,b, 1973). There is no circadian variation in protein synthesis in R15 (Strumwasser and Wilson, 1976; see Section IV. 1). The regulation of biosynthesis of a 12 K precursor in R15 has been suggested to be dependent on synaptic input (Gainer and Barker, 1974). Its synthesis is apparently selectively reduced 30–50% when R15 is inhibited by synaptic input (Gainer and Barker, 1975). However, Aswad (1978) was unable to confirm this observation. A 12 K peptide or a group of protein species of similar molecular weights occurs in several neuronal types in *Aplysia*, not all of them neurosecretory (Wilson, 1974; Berry, 1976a), but the 6–9 K peptides occur only in R15 and other neurones with

neurosecretory granules (Loh and Peterson, 1974; Berry, 1975). The processing (Loh and Gainer, 1975b; Berry, 1976b) and kinetics of movement from the soma to the axon (Berry, 1980) of these proteins have been studied in R15. The latter probably reveal a multiple pool system for storage (Berry, 1980). Furthermore, in R15 and the other putative neurosecretory neurones, the 12 K peptide or peptides (13.1 K in R15; Aswad, 1978) are very likely to be precursors of the 6–9 K peptides (Loh and Gainer, 1975a,b; Strumwasser and Wilson, 1976; Aswad, 1977).

R15 has provided a good model system for studying the regulation of synthesis of peptides from large precursors (Gainer *et al.*, 1977). It is now clear that R15 is also amenable to investigation by molecular genetic means. Kupfermann and Weiss (1976) found that extracts from R15 increased the weight (presumably the water content) of *Aplysia* into which they were injected. The latter observation is consistent with a neurosecretory function and suggests a role in control of water balance by neurohormonal means. Buck *et al.* (1983) have cloned a gene, apparently unique to R15, which they hypothesize codes for a precursor of the R15 water-regulatory peptide.

The evidence that R15 is involved in osmoregulation and water balance comes also from experiments in which the osphradium was subjected to experimental stimuli during intracellular recording from R15. Stinnakre and Tauc (1966, 1969) found that diluted sea water applied to the osphradium markedly inhibited the rhythmic activity of R15. Hyperosmotic sea water had no effect. These findings were confirmed by Jahan-Parwar *et al.* (1969), Bablanian and Treistman (1983a) and by Maat *et al.* (1983) who show some evidence that at least part of the response to osphradial stimulation may be mediated by Interneurone II which is presynaptic to R15 (see Section IV). Input conveying information about changes in tonicity of the sea water bathing the animal appears to be mediated also by Interneurone XIII (Table 2) via the right pleurovisceral connective (Woodson and Schlapfer, 1979). Experimental hyperpolarization of R15 in the intact animal results in changes in haemolymph ion and metabolite concentrations possibly related to an osmoregulatory function (Bablanian and Treistman, 1983b). It remains to be demonstrated that exposure of the osphradium to sea water of differing salinity affects the release by R15 of a neurosecretory substance that is capable of producing osmoregulatory changes.

Until recently, virtually nothing was known about the existence of neurones post-synaptic to R15. Brown and Mayeri (1982) have observed slow synaptic potentials in phase with R15's burst cycle in neurones in the lower left quadrant of the abdominal ganglion. It is possible that the transmitter in this case is a peptide although R15 is not an FMRF-amide immuno reactive neurone (Brown *et al.*, 1984). With regard to conventional neurotransmitters, analysis of enzymes and precursors in R15 provides mainly negative evidence on the possibility of classical synaptic output. The soma of R15 has no detectable choline transferase activity (Giller and Schwartz, 1971a; McCaman and Dewhurst, 1970; Eisenstadt *et al.*, 1973) and virtually no acetylcholine (McCaman *et al.*, 1973; McCaman and McCaman, 1976) and is therefore unlikely to be cholinergic. Although histidine is uniformly present throughout *Aplysia* neurones, including R15 (Weinreich, 1976; Weinreich and Weinreich, 1977), there is no histidine decarboxylase activity in R15 indicating that histamine biosynthesis is unlikely (Weinreich, 1976; Weinreich and Yu, 1977). Octopamine is not detectable in R15 (Saavedra *et al.*, 1974; McCaman and McCaman, 1978) and serotonin occurs at a low level only (Weinreich *et al.*, 1973, 1975; Ono and McCaman, 1984). Significant levels of DOPA decarboxylase have been found in R15, which might indicate metabolic pathways for production of dopamine and/or serotonin (Weinreich *et al.*, 1972). R15 contains comparatively small amounts of glycine and has moderate levels of aspartate (Zeman and Carpenter, 1975) and glutamate (Borys *et al.*, 1973) but probably not enough to support a transmitter function (Iliffe *et al.*, 1977). Only taurine occurs at comparatively high levels (McCaman and Stetzler, 1977). These observations suggest but by no means prove that R15 may be solely a neurosecretory or at least peptide-transmitter producing neurone.

3. *Characteristic Endogenous Bursting Activity*

The property of R15 which has proved to be the most characteristic and of great interest to

electrophysiologists has been its bursting activity of which a typical example is shown in Fig. 1A. The burst consists of a train of action potentials whose frequency increases until midburst and then decreases again. The positive overshoot increases to mid-burst and then remains quite constant, and the undershoot (negative after-potential) decreases almost throughout the burst. At the end of the burst there is a characteristic, slow depolarizing after-potential (DAP) followed by the post-burst hyperpolarization. The hyperpolarization decays during the inter-burst interval at an increasing rate leading to a depolarizing inflection which triggers the first action potential of the next burst.

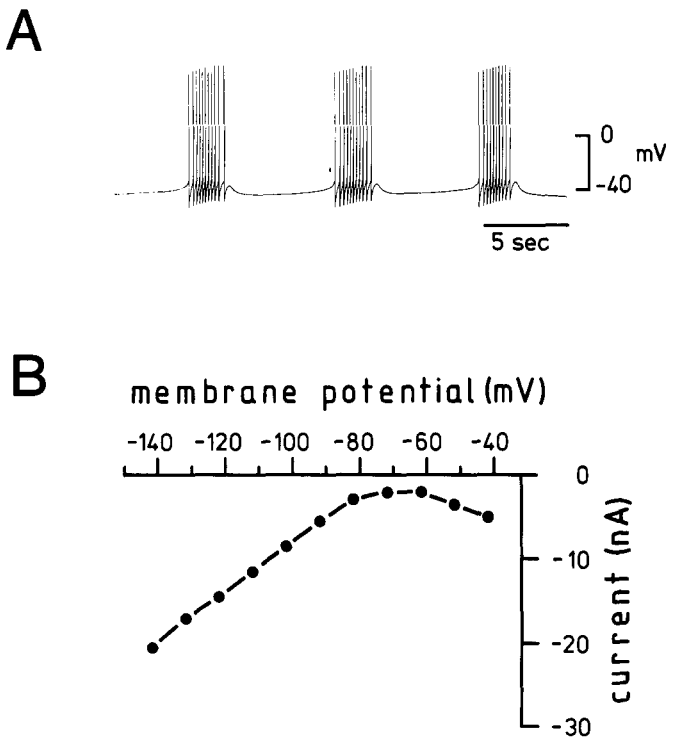


FIG. 1. Characteristic electrical activity of neurone R15.

A. Burst activity recorded from R15 in an isolated ganglion perfused with normal *Aplysia* medium at room temperature. Rhythmic activity in R15 consists of bursts of action potentials separated by inter-burst hyperpolarizations. Some individual variation is seen between animals, particularly in the number of action potentials per burst and the duration of the inter-burst hyperpolarizations, but the general features of the bursts remain unchanged.

B. Steady-state *I–V* curve obtained under voltage-clamp. The current required to hold the cell at each membrane voltage (abscissa) is plotted on the ordinate. In this, and all following illustrations, positive current corresponds to an outward flow of membrane current, i.e. a movement of positive ions from inside the cell to the outside, or a movement of negative ions in the opposite direction. In the absence of pharmacological treatment, the currents measured under voltage clamp generally reflect the sum of many different underlying currents.

This bursting activity is not driven by synaptic input. It is endogenous to the neurone. The evidence for this was initially negative. Arvanitaki and Chalazonitis (1958, 1961b), Strumwasser (1965, 1966) and Frazier *et al.* (1967) were all unable to find evidence of periodic synaptic input. Furthermore, the burst frequency of R15 and other bursting neurones can be altered by constant current injection into the soma (Arvanitaki and Cardot, 1941a; Arvanitaki and Chalazonitis, 1968) and the burst rhythm can also be phase shifted by current injection (Strumwasser, 1967a; Pinsker, 1977a) or a single inhibitory synaptic input (Pinsker, 1977a,b,c). Conclusive evidence for the endogenous origin of rhythmic bursting was provided by Alving (1968) and Chen *et al.* (1971) who recorded from isolated somata, Alving by ligaturing the axon, Chen *et al.* by dissociating the ganglion with trypsin. These isolated

somata exhibited various types of spontaneous activity including bursting in R15 and other unidentified neurones. It appears from the literature that in all cases of bursting in isolated neurones a part of the axon remained in functional contact with the soma. Chen *et al.* obtained three entirely spherical cell bodies which showed regular spontaneous firing but not bursting. R15 appears to lose its bursting properties when cultured (Dagan and Levitan, 1981). We have been unable to obtain rhythmic bursting in isolated R15 somata from which all trace of the axon and axon hillock have been separated. These findings are significant since we will show in Section III that axonal action currents may be the source of an important burst current.

The rhythmic bursting shown in Fig. 1A is typical for an R15 in an isolated abdominal ganglion from an adult animal. It appears that bursting develops in R15 with the growth of the *Aplysia* (Ohmori, 1981). Perhaps it is significant that the formation of a bursting capability coincides with the appearance of cytoplasmic granules (cf. Section II.2). Increased age also affects modulation of R15 bursting via the RC1 synaptic input (Skinner and Peretz, 1983) (Section IV.1), and is accompanied by an accumulation of lipofuscin (Tudor *et al.*, 1979). In the intact *Aplysia* or a dissected intact nervous system, in contrast to the isolated ganglion, the activity of R15 can be highly irregular as the neurone is bombarded with synaptic input (Stinnakre and Tauc, 1969; Woodson and Schlapfer, 1979, Moore *et al.*, 1981).

The next section is an outline of the ionic basis of endogenous bursting and in Section IV we will describe the modulatory effects of synaptic input and neurotransmitters on this rhythmic bursting activity.

III. IONIC BASIS OF BURSTING

1. *The* I–V *Curve and Its Significance*

The cell membrane of R15 contains a variety of voltage-dependent and activity-dependent conductance mechanisms. As in all spiking neurones and many other excitable tissues, some of these conductances are active only at membrane potentials above the threshold for action-potential generation. Others, however, activate and inactivate at more hyperpolarized membrane potentials, including those potentials through which the R15 cell membrane passes during the course of the endogenous burst cycle. The number and complexity of these conductances means that simply recording the membrane potential of the neurone is insufficient for an assessment of the contribution of particular conductances to the neuronal activity. The technique known as "voltage-clamping" has been developed to ameliorate this problem, and the use of this technique has provided many of the data discussed in this section.

In order to voltage-clamp a cell, two electrodes are used. One measures membrane potential and the other passes current. The signal from the voltage measuring electrode is led back to the voltage-clamp as the "feedback" input. A second input to the clamp provides a "command" signal, i.e. the voltage that the experimenter wishes to impress upon the membrane. The measured membrane voltage and the command signal are compared and the difference or "error" signal is amplified and fed to the cell via the current passing electrode to reduce the error. In a properly functioning voltage-clamp, the membrane voltage is controlled by the experimenter and the current that is required to maintain that voltage becomes the signal that is measured.

The relationship between membrane current (measured) and membrane voltage (controlled) can then be plotted as a current–voltage or I–V curve. If the membrane voltage is held at a number of different voltages for long periods of time, or swept very slowly over a range of voltages, then the "steady-state" I–V curve results from plotting the measured current against the corresponding potential. Such a steady-state I–V curve for neurone R15 is illustrated in Fig. 1B. This curve is very nearly linear for potentials more negative than -80 mV (i.e. it has a nearly constant slope conductance, $g_{slope} = \Delta I/\Delta V$). The slope conductance gives a measure of the sum of the various ionic conductances that are active in this region of potentials. It should be noted that slope conductance provides a more meaningful description of the underlying ionic conductances than does the absolute

conductance ($G=I/V$) because the underlying currents rarely change sign at $V=0$. For example, a potassium current carried by voltage-insensitive channels would be described by $I_K=g_K(V-E_K)$ where E_K is the potassium equilibrium potential. When $V=E_K$, the current is zero and when $V>E_K$ the current is positive, or outward. At the equilibrium potential, the ion flux produced by the electrical field across the membrane is exactly balanced by the flux arising from the diffusion gradient. The value of the equilibrium potential thus depends on the ratio of inside and outside (cytoplasmic and bathing medium) ion concentrations, according to the Nernst equation:

$$E_X=\frac{RT}{zF}\ln\frac{[X]_{out}}{[X]_{in}} \tag{1}$$

where X is the permeant ion, R=gas constant, T=absolute temperature, F=Faraday's constant, and z=the valence of the ion, including its sign. At 20°C,

$$E_X\simeq -58\text{ mV}\log_{10}\frac{[X]_{in}}{[X]_{out}} \tag{2}$$

In cell R15, $[K^+]_{in}$ is approximately 200 mM (Kunze and Brown, 1971) and in normal *Aplysia* medium $[K^+]_{out}$ is 10 mM, giving an E_K of approximately −75 mV.

The I–V curve of R15 below −80 mV in fact reflects almost entirely a potassium conductance. If this potassium conductance is blocked by addition of Cs^+ to the bathing medium, the slope of the I–V curve is reduced several fold (Benson and Levitan, 1983). At potentials more positive than −80 mV, the slope becomes less steep, a phenomenon defined as "anomalous rectification" (Katz, 1949). This expression is synonymous with "inward rectification". The term "anomalous" was chosen historically because the well-known axonal I–V curves generally become steeper as the membrane potential is made more positive due to activation of the "delayed rectifier", I_K (Section III.2) (i.e. the slope conductance becomes larger with increasing voltage). In the case of R15, anomalous rectification arises from two sources. First, the potassium conductance that provides most of the membrane current at very negative potentials (> −80 mV, the linear region of the I–V curve) is itself anomalously or inwardly rectifying, displaying an ability to pass large inward but only small outward K^+ current (Eaton *et al.*, 1978; Benson and Levitan, 1983). This conductance thus becomes smaller and non-linear as the membrane potential is made more positive than E_K (−78 mV), and it appears to carry no current at potentials more positive than −40 mV (Drummond *et al.*, 1980a; Benson and Levitan, 1985). This current has been designated I_R (Benson and Levitan, 1982, 1983) and it is discussed further in Section III.2. A second source of anomalous rectification in R15 is the voltage-dependent activation of an inward current. Activation of this current begins at −80 to −60 mV and increases up to −30 to −10 mV (Smith *et al.*, 1975). The increasing activation with increasing voltage leads to a larger and larger inward, or negative, current flow. Thus, at potentials more positive than about −60 mV, the membrane current becomes more negative as the voltage is made more positive. Another way of describing this relationship is to describe the I–V curve in this range of potentials as a region of "negative slope conductance" or, to follow historical usage, "negative slope resistance" (NSR). The inward current responsible for the NSR we refer to below as I_{NSR} (Section III.2).

The voltage dependence of I_{NSR} gives rise to a situation that may at first seem paradoxical: an *increase* in membrane slope conductance can be produced by a *decrease* in the conductance responsible for I_{NSR}. The reason behind this is, of course, the voltage dependence of I_{NSR} which gives it a negative slope conductance. Since the slope conductance of the membrane at a particular voltage is simply the arithmetical sum of all the ionic slope conductances that are active at that voltage, reducing the size of the (negative) I_{NSR} conductance leads to a net (positive) increase in the membrane slope conductance. The inverse dependence of I_{NSR} and g_{slope} appears to explain an early and oft-repeated finding in R15, that the membrane slope conductance decreases through the interburst interval (Carpenter, 1973; Junge and Stephens, 1973; Arvanitaki and Chalazonitis, 1968). This

finding has been taken as support for the idea that the interburst hyperpolarization is mediated by a K^+ conductance that is activated during the burst and which then decays during the interburst. Recent observations strongly suggest that in R15 the current which mediates the interburst is not a K^+ current (Adams and Levitan, 1985; Kramer and Zucker, 1985b) but, rather, arises from an inactivation of I_{NSR}. The observed decrease in slope conductance during the interburst thus results from a reactivation of I_{NSR}. It is important to bear in mind that different mechanisms may occur in other bursting neurones.

Although the I–V curve in Fig. 1B has been referred to as a steady-state I–V curve, it is a steady-state that is imposed by the voltage-clamp, not one produced by the cell itself. This can be inferred from the position of the I–V curve with respect to the abscissa or zero-current axis. For all voltages up to and including the threshold for action potential generation (usually -30 to -20 mV in this cell), the I–V curve lies below the abscissa. This means that there is a continual negative, or inward, flow of current. If the voltage-clamp were turned off, this inward current flow would depolarize the cell, leading to action potential generation. In order for a cell to be stable in the absence of any externally applied current, its I–V curve must pass through the zero-current axis with positive slope. If this requirement is met, then the membrane potential of the cell will remain stable at the potential where the I–V curve intersects the abscissa (i.e. where $I=0$). It should be noted from Fig. 1B that the I–V curve approaches to within a few nA of the abscissa at potentials between -80 and -60 mV. Thus, a small increase in outward current would be sufficient to bring about stability in the cell (i.e. to inhibit its endogenous activity). The implications of this observation in relation to modulation of bursting will be considered in detail in Section IV.3.

The region of NSR in R15 has long been associated with the production of bursting activity in this cell (Wilson and Wachtel, 1974). Treatments, such as cooling (Wilson and Wachtel, 1974), application of dopamine (Wilson and Wachtel, 1978), or activation of synaptic input (Adams *et al.*, 1980b), which abolish the NSR in the steady-state I–V curve also inhibit bursting in an unclamped cell. Thus, it appears that the association is more than simply coincidental. However, an NSR is also found in the I–V curves of many non-bursting cells (Partridge *et al.*, 1979). Therefore, even if the NSR is necessary for bursting, it is not sufficient. It may be that it is not the NSR *per se* that is important, but the fact that the NSR keeps the I–V curve below the abscissa and, hence, reflects a continual depolarizing current flow that takes the membrane potential up to the threshold for action potential generation.

In addition to steady-state I–V curves, voltage-clamp techniques are often used to measure the voltage-dependence of many transient currents. One of the earliest studies making use of this technique is that of Hodgkin and Huxley on the squid giant axon (Hodgkin *et al.*, 1952; Hodgkin and Huxley, 1952a,b,c,d). They found that if the membrane potential were held near the resting potential and stepped to more positive potentials, then two distinct currents could be identified by their kinetics. The faster of the two is an inward current, carried by Na^+, which inactivates with time. The second, slower current, which on the time-scales of their experiments remained as a steady-state outward current for long pulses, is carried by K^+. These two currents, together with a voltage-independent leak conductance, were sufficient to account for their data. Furthermore, by changing the ionic medium (Hodgkin and Huxley, 1952a), or by making use of specific pharmacological blockers discovered since the time of those experiments, I_{Na} and I_K can be studied quite independently of one another. The situation is more problematic when using voltage steps to study the I–V curve of a voltage-clamped cell body. In the range of potentials between -100 and 0 mV, some nine or more currents can be identified (see Section III.2) that are carried by Na^+, Ca^{2+} and K^+. These currents have differing kinetics and differing pharmacological sensitivities, but there is often enough overlap between them to make complete separation difficult. As a result, I–V curves that are obtained using step-changes in membrane potential often measure the effects of a large number of different currents.

In the case of R15, the currents that contribute to generation of bursting activity have kinetic time constants that range between milliseconds and tens of seconds and so the measured currents depend not only on the measurement voltage and the time at which the measurement is made, but also on the conditions prior to the voltage clamp pulse. One

example of this, the difference between $I-V$ curves obtained by holding the cell under voltage-clamp, and those obtained by turning on a voltage-clamp when the cell is bursting, is discussed further in Section III.3.

2. *Measured Membrane Currents*

Some neuronal membrane currents have very similar or identical properties over a wide range of vertebrate and invertebrate species (e.g. the TTX-sensitive sodium current I_{Na}). However, the nomenclature of ionic currents presents considerable difficulties due to the ambiguities which arise from comparison of one identified neurone with another in the same species, homologous neurones between different species, or neurones with muscle (cf. recent reviews by Edwards, 1982, and Latorre and Miller, 1983). In Table 1 and Fig. 2, we have presented the major currents measurable during voltage clamp of the soma of R15 utilizing, and extending where necessary, the notation of Adams *et al.* (1980a) in their survey of ionic currents in molluscan neuronal somata. Other important surveys of ionic currents are given by Reuter (1984) and Noble (1984) for cardiac muscle, and Crill and Schwindt (1983) for mammalian central neurones. Undoubtedly some channel types are common to all varieties of excitable cells but others differ so markedly in their properties, despite carrying the same ionic species, that they must represent distinct channel types and require different notation. In the following summary of ionic currents, we have confined ourselves as far as possible to the properties of the currents measured in R15 itself (Fig. 2), with occasional comparison of similar currents in other molluscan neurones. This inevitably means that a number of publications dealing with aspects of molluscan ionic currents which are not directly relevant to burst production in R15 have not been discussed. On the other hand, we have been able to base the model described in Part 3 of this section solely on the currents measured in the neurone whose activity we are attempting to explain. The details of the kinetics and voltage-dependence are given with the model and extensive comparison of some of these currents measured in neurones of different molluscan species is made by Adams *et al.* (1980a). Sodium and calcium gating currents recorded in R15 are described by Adams and Gage (1976, 1979c), and the effects of various ions on bursting *per se* have been investigated by Barker and Gainer (1975a,b).

(a) *Inward currents*

R15 has four inward currents with remarkably diverse kinetics. I_{Na}, a *fast sodium current*, is the classical TTX-sensitive, inward current contributing to the depolarizing phase of the somal action potential. In R2, the *Aplysia* giant neurone, the somal action potential includes contributions from I_{Na} and the fast calcium current, I_{Ca} (Geduldig and Junge, 1968; Geduldig and Gruener, 1970) and the same appears to be the case for the R2 axonal action potential (Horn, 1977, 1978). In R15, the somal action potential is constituted similarly to that of R2 (Carpenter and Gunn, 1970; Adams and Gage, 1979a,b; Gorman and Hermann, 1982) but although the axonal action potential is blocked by TTX it is not known whether it has a Ca^{2+}-dependent component in R15.

I_{Ca}, a *fast calcium current* with kinetics slower than those of I_{Na}, is activated at positive potentials and contributes to the somal action potential. The Ca^{2+} entering the neurone during an action potential or similar depolarization under clamped conditions has been shown by aequorin and Arsenazo III studies to accumulate during repetitive depolarization and to dissipate at a slower rate (Stinnakre and Tauc, 1973; Thomas and Gorman, 1977; Smith and Zucker, 1980; Gorman and Thomas, 1977, 1978, 1980a; Gorman *et al.*, 1984). Inactivation of I_{Ca} has been shown to depend on accumulation of Ca^{2+} near the inner surface of the soma membrane in R15 (Tillotson and Horn, 1978; Tillotson, 1979; Eckert and Tillotson, 1981; Eckert and Ewald, 1982). Not surprisingly, this is also the location of the maximum Ca^{2+}-buffering capacity of R15 (Tillotson and Gorman, 1980, 1983; Gorman *et al.*, 1984). In *Helix* neurons, intracellular Ca^{2+} accumulation also depresses the subsequent activation of K^+ efflux (I_C) (Eckert and Lux, 1977). The I_{Ca} channels conduct Ba^{2+} (Gola and Ducreux, 1977) and their other properties also conform to the general characteristics of "calcium channels" (reviewed by Hagiwara and Byerly, 1981).

TABLE 1.

Letter Code	Current	Activation Range	Kinetics	Blocked by	Not Blocked by	References
Na	Fast, transient, sodium current	−20 mV and positive	Fast	TTX		Geduldig and Junge (1968), Geduldig and Gruener (1970), Connor (1977), Adams and Gage (1979a,b)
Ca	Fast calcium current	−10 mV and positive	Fast	Co^{2+}, Cd^{2+}, Mn^{2+}, Ni^{2+}, La^{3+}	TTX	Geduldig and Junge (1968), Connor (1977), Adams and Gage (1979a,b), Stinnakre and Tauc (1973)
NSR	Current responsible for negative slope region	−70 mV and positive	Slow	?		Wilson and Wachtel (1974), Smith *et al.* (1975), Gola (1974), Gorman *et al.* (1982a), Colmers *et al.* (1982)
D	Transient, depolarizing after-current	−20 mV and positive	Slow	Ca-blockers, low sodium		Adams and Levitan (1982a, 1985), Lewis (1984), Adams (1985), Kramer and Zucker (1985a)
K	"Delayed" potassium current (delayed rectifier)	−15 mV and positive	Intermediate	TEA, 4AP	Cs^{+}, Co^{2+}	Connor and Stevens (1971a), Neher and Lux (1972), Adams and Gage (1980), Hermann and Gorman (1981a,b)
C	Calcium-activated potassium current	0 mV and positive	Intermediate	TEA, Ca-blockers	Cs^{+}, 4AP	Meech and Strumwasser (1970), Meech (1972, 1978), Gorman and Thomas (1980b), Hermann and Gorman (1981a,b)
A	Fast, transient potassium current	−70 mV and positive	Fast	4AP	TEA, Cs^{+}	Thompson (1977), Connor and Stevens (1971b), Nehr (1971), Byrne (1980b), Shimahara (1981, 1983)
H	Transient, hyperpolarizing after-current	−20 mV and positive	Very slow	Ca-blockers, intracellular EGTA	TEA, Cs^{+}, 4AP	Adams (1985), Adams and Levitan (1981, 1985), Kramer and Zucker (1983, 1985b)
R	Anomalously (inwardly) rectifying potassium current	−40 mV and negative	Very fast?	Cs^{+}, Ba^{2+}, Rb^{+}, 4AP	TEA, Tl^{+}	Benson and Levitan (1982, 1983, 1985), Drummond *et al.* (1980a), Benson *et al.* (1983)

Abbreviations: 4AP, 4-aminopyridine; TTX, tetrodotoxin; TEA, tetraethylammonium; EGTA, ethyleneglycol-*bis*-(β-amino-ethyl ether) *N,N'*-tetra-acetic acid.

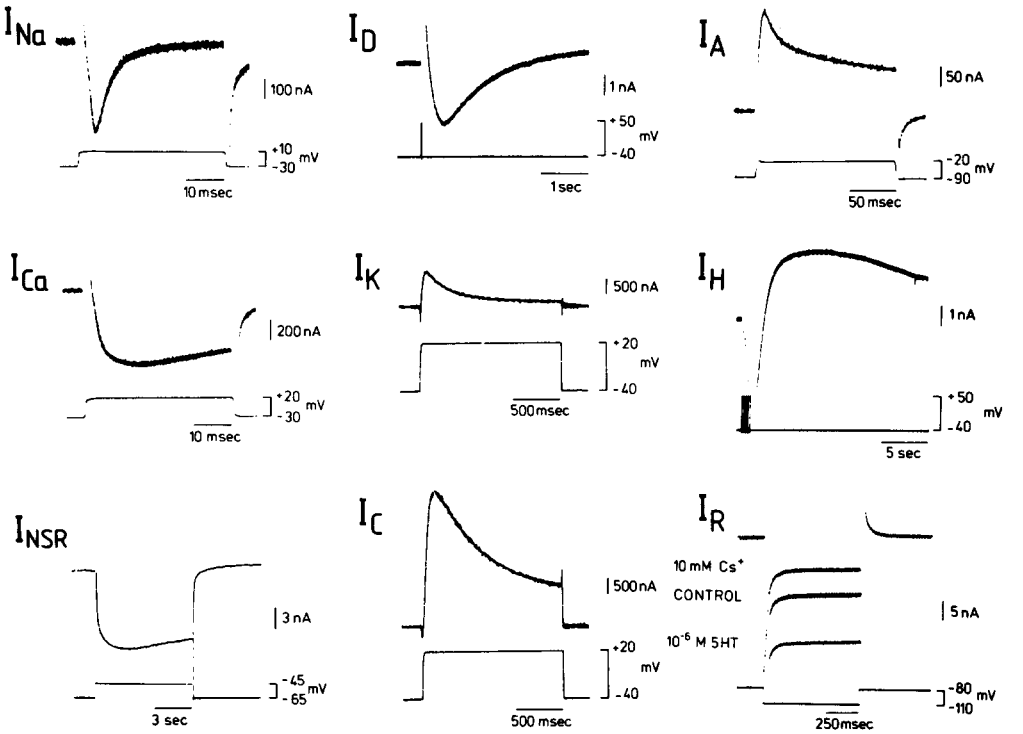


FIG. 2. Individual currents recorded in R15 in response to step or pulse changes in membrane potential. The currents were isolated by a combination of pharmacological treatments and choice of membrane potentials. Please note that the different currents require different time- and amplitute-scales for their illustration and in this figure there are 500:1 ranges in both time and amplitude. A summary of the properties of these currents is given in Table 1, and expanded descriptions are found in the text. I_{Na} fast Na^+ current, fast activation is followed by inactivation. I_{Ca} was blocked by addition of Co^{2+} to the bathing medium, I_K and I_C by TEA and elevation of $[K^+]_o$. I_{Ca} fast Ca^{2+} current, fast activation is followed by slow inactivation. I_{Na} was blocked by addition of TTX to the bathing medium and by lowering $[Na^+]_o$, I_K and I_C by TEA and elevation of $[K^+]_o$. I_{NSR} current responsible for the negative slope region, it consists of a component that is fast to activate and deactivate and a second component that activates and deactivates with time constants of seconds. I_D transient depolarizing after-current. Following action potentials or brief depolarizing pulses, the inward current, I_D peaks at about 400 msec after the pulse and decays with time constants of hundreds of milliseconds. I_K delayed rectifier K current and I_C calcium-activated K current. Both I_K and I_C are voltage-dependent, but I_C also requires Ca^{2+}. The two currents were separated by addition of Co^{2+} to the bathing medium to block Ca^{2+} influx. I_C was measured as the difference between the current in normal medium and that in Co^{2+} medium, I_K as the current that persisted in Co^{2+} medium. I_A fast transient K^+ current. I_A is inactivated by holding the membrane potential above -40 mV; conversely I_K and I_C do not activate below -15 mV. Thus I_A can be studied by delivering depolarizing steps from very negative holding potentials. I_H transient hyperpolarizing aftercurrent. In addition to activation of I_D, action potentials or depolarizing pulses activate a smaller, slower outward current which, because of its slow kinetics, shows temporal summation over long periods of time. I_H in this illustration was elicited by a train of five depolarizing pulses. I_R anomalously rectifying K^+ current. I_R inactivates above -60 mV and activates and deactivates very quickly at more negative potentials. It is inhibited by Cs^+ and enhanced by the addition of serotonin (5HT) to the bathing medium.

I_{NSR} is the current responsible for the *negative slope region* of the I–V curve (Section III.1). As shown in Fig. 2, this current appears to have two components in R15. The fast component has been reported for both bursting and non-bursting neurones, but the slowly activating and inactivating component seems to be characteristic of bursting cells only (Alving, 1969; Partridge *et al.*, 1979). I_{NSR} is an inward current activated by depolarizing steps at sub-threshold membrane potentials. In R15, under physiological conditions, I_{NSR} may be carried by Na^+, by Ca^{2+} or perhaps by both ions. The reason for the uncertainty in determining the charge carrier is a chaos of conflicting observations and interpretations. It has been reported that the NSR is abolished either by removal of Na^+ (Barker and Smith, 1978; Wilson and Wachtel, 1978; Carnevale and Wachtel, 1980) or by removal of Ca^{2+} (Gorman *et al.*, 1982a)

from the bathing medium. Similar findings have been reported for other neurons. Gola (1974) and Eckert and Lux (1975) blocked the NSR in the *I–V* curves of *Helix* neurones by removing Ca^{2+} from the bathing medium, but Drake and Treistman (1980a) report an enhanced NSR in R15 in response to the same treatment. Smith *et al.* (1975) blocked the NSR in cell 11 of *Otala* by removal of Na^+. We have found that complete removal of either Na^+ or Ca^{2+} from the bathing saline results in highly abnormal changes in membrane resistance. Futamachi and Smith (1982) reported that fresh TTX blocks I_{NSR} in R15, and Johnston (1976) concluded that I_{NSR} can be carried by both Na^+ and Ca^{2+} ions in neurones L_2–L_6 of *Aplysia*. In Cs^+-loaded cells R2 and LP1 of *Aplysia*, I_{NSR} is carried by Na^+ ions, and much of its apparent inactivation in the normal neurone is due to slow activation of an outward current that is blocked by intracellular Cs^+ (Colmers *et al.*, 1982). Heyer and Lux (1976), recording from *Helix* neurones, also found that the slow inward current occurs simultaneously with an efflux of K^+ ions.

I_{NSR} is highly temperature dependent (Wilson and Wachtel, 1974; Eaton and Brodwick, 1976) and this property probably contributes to the temperature-dependence of the activity pattern in R15 and other molluscan bursting neurones (Chalazonitis, 1961; Murray, 1966; Carpenter, 1967, 1981; Strumwasser, 1973; Beiswanger and Jacklet, 1975; Moffett and Wachtel, 1976; Benson and Jacklet, 1977; Wiederhold and Carpenter, 1982; Treistman and Bablanian, 1983). Pinsker (1977a) noticed a hysteresis effect in the response of the burster L3 to changes in temperature. It would be interesting to know whether individual currents such as I_{NSR} exhibit the same effect.

Eckert and Lux (1975, 1976) and Gola (1974) described a slowly activating inward Ca^{2+} current characterized by extremely slow decay and a voltage dependence similar to I_{NSR}. This current may be similar to the slow component of I_{NSR} in R15, and it has been tentatively identified with the slow, Ca-dependent tail current, I_B, recorded in some molluscan bursters (Adams *et al.*, 1980a). I_B is not seen in non-bursting neurones whether or not they exhibit the fast phase of I_{NSR}. For left upper quadrant bursting (LUQB) neurones, Kramer and Zucker (1984) conclude that I_B contributes to the DAP (i.e. I_D, defined below) and that it has properties which make it distinct from I_{NSR}. However, the current state of the evidence for R15 suggests that the slow tail current I_B contributes to I_{NSR} and may be conducted by the same channels as the slow component of I_{NSR}. The fast component of I_{NSR} may be equivalent to the persistent inward current incorporated into earlier models of bursting in R15. Since I_{NSR} can be recorded from isolated somata, its ion channels must occur in the somal membrane (Partridge *et al.*, 1979). This does not exclude the existence of similar channels in the axon. Indeed, Lewis and Wilson (1983) present evidence that the dopamine-sensitive I_{NSR} channels (see Section IV.3) may be confined to the proximal axon.

I_D, the *transient, depolarizing after-current*, drives the burst after the initiation of the first action potential, and underlies the DAP. It has been described recently in intact R15 (DAC of Lewis, 1984; Lewis *et al.*, 1985) and in axotomized LUQB neurones (L2-L6) of the *Aplysia* abdominal ganglion (Phase II current of Kramer and Zucker, 1985a). I_D is an inward current between -20 mV and -120 mV (Adams and Levitan, 1985; Kramer and Zucker, 1985a) with slow kinetics, decaying over the course of hundreds of milliseconds (Adams, 1985). I_D is blocked by EGTA injection (Lewis, 1984; Kramer and Zucker, 1985a), is blocked by divalent cations which prevent Ca^{2+} influx (Adams and Levitan, 1985), and is activated by Ca^{2+} injection (Kramer and Zucker, 1985a). Hofmeier and Lux (1981) found in *Helix* neurones that Ca^{2+} injection activates not only I_C but also an inward current specific for neither Na^+ nor Ca^{2+}, and they suggested that this inward current contributes to the increase in spike frequency during the early part of the burst. I_D is reduced by decreasing $[Na^+]_0$ (Lewis, 1984; Adams and Levitan, 1985; Kramer and Zucker, 1985a) but is not blocked by TTX (Lewis, 1984). Both cooling (Lewis, 1984) and dopamine (Lewis, 1984; W. B. Adams, unpublished observations) block I_D. Adams and Levitan (1985) find that I_D often displays a sharp activation threshold, particularly when the K^+ currents are blocked. Lewis (1984), on the other hand, reports graded activation. So far it has not been possible to activate I_D in R15 when the axon has been ligated or cut close to the cell body (Adams, 1985; Kramer and Zucker, 1985a). The latter finding does not apply to the LUQB neurones (Kramer and

Zucker; 1985a). Pseudo-bursting observed in a few axotomized neurones by Chen *et al.* (1973a) lacked many characteristics of normal bursting, including the DAP.

It is clear that I_D is activated by action potentials in a bursting R15, and there are at least two ways in which this could occur. Lewis (1984) has suggested that I_D is carried by its own voltage-dependent channels and that it is probably activated by increased intracellular Ca^{2+}. Kramer and Zucker (1984, 1985a) conclude that I_D is probably identical to I_B, but not to I_{NSR}, and that it is therefore a calcium-activated inward current. Adams has suggested that I_D may not be conducted by ion channels in the cell body, but that it is the result of electrotonic spread of current from action potentials propagating along the axon away from the cell body (Adams, 1985; Adams and Levitan, 1981, 1982a, 1985). This requires that the action potentials in the axon hillock be partially Ca^{2+} dependent, and it would account for the unusual ionic dependence of the DAP (Thompson and Smith, 1976; Bulloch and Willows, 1981). Repeated action potentials cause a summation of I_D that is consistent with the conception of a current of electrotonic origin. In LUQB neurones, I_B may play the same role in burst production as I_D does in R15. The details of this role are given in Section III.3.

A Cl^- current activated by hyperpolarization has been described for some *Aplysia* neurones (Chesnoy-Marchais, 1982, 1983), but to our knowledge no experiments have been carried out to determine whether a similar inward current exists in R15.

(b) *Outward currents*

Five outward currents have been described for R15. Four of them are K^+ currents, and one, I_H, is a calcium-mediated current.

I_K, the *delayed rectifier*, is the classical K^+ current contributing to action potential repolarization (Neher and Lux, 1972; Hermann and Gorman, 1979a,b, 1981a,b).

I_C, the *calcium-activated K^+ current*, is activated by an influx of Ca^{2+} ions (Meech and Strumwasser, 1970; Meech, 1972, 1976, 1978; Gorman and Thomas, 1980b; Woolum and Gorman, 1981; Gorman *et al.*, 1982a; Hermann and Hartung, 1983). The I_C conductance appears to be greater in R15 than in the non-bursting neurone L11 (Gorman and Hermann, 1982). I_C can be blocked or reduced by caffeine (Hermann, 1981), quinidine (Hermann and Gorman, 1984), and by various divalent cations, some of which (e.g. Ba^{2+}) can enter the cell through Ca^{2+} channels (Hermann and Gorman, 1979b; Gorman and Hermann, 1979). Thompson (1977), using *Tritonia* neurones, was able to differentially block I_K with extracellular TEA and I_C with extracellular Co^{2+} or Mn^{2+} (which blocked Ca^{2+} influx), and found that 4-aminopyridine (4AP) did not block I_K or I_C. Applied to R15, tetraethylammonium (TEA) blocks I_K and I_C (Meech, 1974b), and it appears to block I_C more effectively (Hermann and Gorman, 1979a, 1981b). We found it difficult to separate the two currents, I_K and I_C, which have similar kinetics and voltage-dependence, using TEA. To obtain a record of I_K we measured the total outward current and subtracted the component (I_C) blocked by Co^{2+} (Fig. 2). In R15, TEA also appears to have direct effects on intracellular Ca^{2+} accumulation (Zucker and Smith, 1979). In Part 3 of this Section, we will argue that I_C is *not* responsible for the post-burst hyperpolarization in the R15 burst cycle, contrary to most hypotheses (Junge and Stephens, 1973; Meech, 1974a; Smith *et al.*, 1975; Kandel, 1976; Gulrajani and Roberge, 1978; Smith, 1980; Johnston, 1980; Carnevale and Wachtel, 1980; Gorman *et al.*, 1980, 1981, 1982a). The post-burst hyperpolarization is produced by I_H.

I_A, the *fast transient K^+ current*, is activated at hyperpolarized potentials in many molluscan neurones (Hagiwara and Saito, 1959; Stevens, 1969; Neher, 1971; Connor and Stevens, 1971b; Thompson, 1977) and inactivated in R15 between -70 and -50 mV (Benson and Levitan, 1983). I_A is blocked by 4AP but not by extracellular TEA (Neher and Lux, 1972; Thompson, 1977, 1982). If I_A has a role in the bursting activity of R15, this role is unknown, although it has been suggested that I_A contributes to interspike delay in beater (Connor and Stevens, 1971b,c; Connor, 1978, 1982) and burster neurones (Klee and Faber, 1974). I_A may have a role in presynaptic modulation of synaptic output in *Aplysia* (Shimahara, 1981, 1983), and in the integrative action of motor cells (Byrne, 1980a,b).

I_H is a *transient, hyperpolarizing after-current* which arises from a decrease in inward Ca^{2+} current and which produces the post-burst hyperpolarization in R15. It has been described

recently in R15 (HAC of Lewis, 1984; Adams, 1985; Adams and Levitan, 1981, 1985) and in the LUQB neurones L2-L6 of *Aplysia* (Phase III current of Kramer and Zucker, 1983, 1985b). I_H is activated in a very graded fashion (Adams and Levitan, 1985) and it is an outward current at membrane potentials between -20 mV and -120 mV (Adams and Levitan, 1985; Kramer and Zucker, 1985b). Divalent cations which block Ca^{2+} currents also block I_H (Adams and Levitan, 1985; Kramer and Zucker, 1985b). I_H is dependent on $[Ca^{2+}]_0$ (Adams and Levitan, 1985) and it is activated less under voltage-clamp by voltage commands which approach the calcium equilibrium potential, E_{Ca} (Kramer and Zucker, 1985b). I_H is blocked by injection of EGTA (Lewis, 1984) and by dopamine (W. B. Adams unpublished observations; Lewis, 1984). The K^+-dependence of I_H is unclear. Adams and Levitan (1985) see no K^+-dependence, while a small K^+-dependence is reported by Lewis (1984) and Kramer and Zucker (1985b). I_H does not reverse at E_K (Adams and Levitan, 1985; Kramer and Zucker, 1985b), and it is independent of $[Na^+]_0$ (Adams and Levitan, 1985). Prolonged exposure to zero $[Na^+]_0$ blocks I_H (Lewis, 1984; Adams and Levitan, 1985; Kramer and Zucker, 1985b) but we suspect that absence of extracellular Na^+ causes non-specific and deleterious effects on *Aplysia* neurones, affecting the properties of many currents. It is possible that I_H and one or both components of I_{NSR} are mediated by the same channels since the two currents have reciprocal voltage-dependence within the range tested (-80 mV to -30 mV) and both are reduced by dopamine (Adams and Levitan, 1985, unpublished observations). Calcium-dependent inactivation appears to be responsible for many properties of both currents. The evidence that I_H produces the post-burst hyperpolarization in R15 is given in Part 3 of this section.

I_R, the *anomalously (inwardly) rectifying K^+ current*, is similar in most of its properties to the anomalous rectifiers described for muscle (Eaton and Brodwick, 1976; Benson and Levitan, 1982, 1983, 1985). Although it is possible that I_A might be responsible for some anomalous rectification in molluscan bursters (Gola and Romey, 1971; Connor and Stevens, 1971b, Neher, 1971), I_R is a separate current conducted by channels distinct from those of I_A (Benson and Levitan, 1983). Since it is an imperfect inward rectifier, I_R contributes a small outward current at potentials 0 to 20 mV positive to the potassium equilibrium potential ($E_K = -78$ mV), and it increases in response to serotonin. The role of I_R in burst modulation will be discussed in detail in Section IV.3.

(c) *Metabolic pump currents*

Many studies have shown the presence of a metabolic Na^+-K^+ pump in the neurones of *Aplysia* and other molluscs (reviews by Thomas, 1972; Carpenter and Gregg, 1984). At least half of the actively extruded Na^+ is electrogenic, contributing to the membrane potential (Carpenter and Alving, 1968; Carpenter, 1970; Cooke *et al.*, 1974). Some experiments involving pump inhibition by low temperature present problems of interpretation because of the great temperature sensitivity of I_{NSR} which is activated at subthreshold potentials (e.g. Willis *et al.*, 1974; Junge and Ortiz, 1978). Changes have been reported in the pattern of rhythmicity in R15 in response to experimental manipulation of P_{O_2}, $[K^+]_0$ and $[Na^+]_0$, and to the application of ouabain (Chalazonitis, 1963; Arvanitaki and Chalazonitis, 1968; Meunier and Tauc, 1970; Carpenter, 1970; Steffin, 1975; Coyer, 1977; Coyer *et al.*, 1981). There is, however, no compelling evidence supporting earlier hypotheses that a metabolic pump current is in large part responsible for rhythmicity or the post-burst hyperpolarization (Strumwasser, 1965; Junge and Stephens, 1973; Chen *et al.*, 1973a,b).

Pinsker and Kandel (1969) and Sawada *et al.* (1980) have suggested that synaptic activation by L10 of an electrogenic Na^+ pump current contributes to the inhibition of long duration (ILD) recorded in neurones L3-L6, and three ventral, right caudal abdominal ganglion neurones of *Aplysia*. ILD in R15 is insensitive to treatment with ouabain (Parnas and Strumwasser, 1974; cf. Ascher, 1968) as is the response of R15 to serotonin (Drummond *et al.*, 1980a). The evidence in favour of a synaptic mechanism involving pump activation in *Aplysia*, particularly with reference to ILD, has been re-evaluated by Kehoe and Ascher (1970) and we concur with their conclusion that the case remains to be proven.

For present purposes, we have assumed that in R15 the Na^+-K^+ pump makes a

contribution to the resting potential which is essentially unaffected by changes in $[K^+]_0$ and other parameters during the course of the burst cycle.

3. *Ionic Model of Bursting with Computer Simulation*

The remainder of Section III concerns the details of our present conception of how the membrane currents of R15 might interact to produce the observed endogenous, rhythmic bursting activity. The accompanying computer simulation utilizes the activation and inactivation kinetics, voltage sensitivities and amplitudes, so far as these parameters have been determined, of the currents measured in R15. It is thus purely a refinement of the physiological model proposed here and depends for its validity upon the quality of the available electrophysiological data derived from R15. In the course of explaining our hypothesis, we will discuss, where appropriate, earlier hypotheses for bursting in relation to our proposed mechanism. However, it is worthwhile to begin with a brief summary of previous models of R15's activity, including electronic and purely mathematical models.

There have been numerous reviews of the ionic mechanisms of molluscan bursters. Those which deal explicitly with R15 include the comprehensive surveys of biological oscillators by Rapp (1979) and Berridge and Rapp (1979). The hypotheses of Waziri *et al.* (1965), Mathieu and Roberge (1971), Carpenter (1973), Wachtel and Carnevale (1974), Klee and Faber (1974), Kandel (1976), Moffett (1977), Gola (1974, 1978) and Barker and Smith (1978) are representative of the early qualitative models based on an unstable balance of opposed ion currents, most of them not precisely identified with physiologically measurable currents, and emphasizing the importance of the NSR or anomalous rectification in the I–V curve of R15. The source of the post-burst hyperpolarization was usually assumed to be I_C (e.g. Meech, 1974a) and this hypothesis has been explicitly formulated in some detail by a number of authors and will be discussed below. Kandel (1976) gives a synthesis of the data and models of Faber and Klee (1972), Gola (1974), Wachtel and Wilson (1973), Smith *et al.* (1975) and Mathieu *et al.* (1976) which can be summarized as a combination of conventional action potential currents, anomalous rectification, a slow $g_K(I_C)$ and slow and persistent g_{Na}'s. This hypothesis was refined by Gulrajani and Roberge (1978) who simulated their concept with an electronic model (Gulrajani *et al.*, 1977; Roberge *et al.*, 1978) and it has been most recently discussed with new data and further refinements by Gorman *et al.* (1980, 1981, 1982a), Wilson (1982) and Lewis *et al.* (1985). Daunicht (1976) presents a model unique in its eccentricity: it denies the existence of ion channels entirely.

Purely mathematical simulations of bursting in R15 have been carried out by Both *et al.* (1976), Plant and Kim (1975, 1976), Plant (1978, 1981), and Mironov (1983). None of these models differs significantly from the traditional one which we will show in the following paragraphs is not the best explanation for bursting based on the experimental data now available for R15.

(a) *Quantitative description of action potential currents*

Conceptually, at least, the components of our model of bursting in R15 can be separated into two groups, those components that participate in action potential generation and those that are active at membrane voltages below the action potential threshold. For the description of currents flowing during the generation of action potentials, we have adopted the formulation used by Hodgkin and Huxley (1952d) in their classic study of the squid giant axon. This approach is outlined below. In the following section we describe how these equations were modified in order to provide a more accurate simulation of action potential generation in neurone R15. Finally, the addition of I_D and I_H, and a set of small subthreshold currents, leads to a model that simulates the characteristic bursting activity of neurone R15.

The form of the equations used by Hodgkin and Huxley followed directly from a number of assumptions and conventions. (In the following discussion, some of the sign conventions have been reversed to be consistent with present usage.)

(1) The current carried by a particular ion species is determined by a specific conductance system (or systems) for that ion, and the conductance systems are independent of each

other. In the 30 years since Hodgkin and Huxley's work, neurophysiologists have grown to accept the concept that ions cross cell membranes via specific channels through the membrane. This view has been supported by the discovery of specific "channel-blocking" pharmacological agents, by the development of techniques that allow the measurement of currents through single channels, and by the chemical isolation of proteins which, when reinserted into artificial lipid membranes, permit ions to cross the membrane. The ion specificity of these channels is not complete, but shows a selectivity, for example, of Na^+ over K^+, that can be quite high.

(2) Quantitatively, the current carried by an ion species, e.g. potassium, can be expressed by

$$I_K(V, t) = g_K(V, t)(V - E_K) \tag{3}$$

where I_K is the K^+ current that flows out of the cell, g_K is the voltage- and time-dependent membrane conductance for K^+, V is the voltage across the membrane (inside of the cell measured with respect to the outside) and E_K is the K^+ equilibrium potential [eqn (2)]. Since I_K, V and E_K are measurable quantities, eqn (3) essentially provides a definition of g_K.

(3) The conductance can be written as the product of a maximal conductance, an activation term, and an inactivation term

$$g_K(V, t) = g_K(\max)\,[\text{activation}]\,[\text{inactivation}] \tag{4}$$

The activation and inactivation terms take on values between 0 and 1 as the result of the recent voltage history of the membrane. Studies of single membrane channels have revealed that channels may take on a number of different "states", presumably conformational states, but from the point of view of allowing ions to pass through them, they are either "open" or "closed". Moreover, the conductance of an open channel is at most mildly dependent on membrane voltage, so that in eqn (4), $g_K(V, t) = g_K(\max)$ when all of the channels are open. The product of the [activation] [inactivation] terms gives the proportion of channels that are open under a particular set of conditions. In the squid axon, Na^+ channels both activate and inactivate in response to membrane depolarization. In contrast, K^+ channels show very little inactivation, so that the last term in eqn (4), [inactivation] = 1. It is useful to introduce at this time the terms "deactivate" and "deactivation". Deactivation is the reverse of the activation process. If the membrane is held at a voltage where activation is small and then stepped to a voltage (usually more positive) where activation is greater, the channels open and the current activates. When the voltage is stepped back to the original potential, the channels close and the current deactivates. Inactivation is a quite separate process that often occurs at the same voltages that cause the channels to open. Inactivation may proceed simultaneously with activation.

(4) Hodgkin and Huxley noticed that activation of K^+ current takes place with a delay, or with sigmoidal, nonexponential kinetics. In contrast, deactivation begins immediately and follows an exponential time course. In order to account for this difference when fitting their data to mathematical expressions (in part, to facilitate the later reconstruction of action potentials) they made an inspired choice. They assumed (a) an underlying process that followed mass-action kinetics, e.g.

$$n_{\text{deactivated}} \underset{k_{-1}(V)}{\overset{k_1(V)}{\rightleftharpoons}} n_{\text{active}} \tag{5}$$

where $k_1(V)$ and $k_{-1}(V)$ are functions of membrane voltage, and (b) that the state of activation depended upon n raised to some power, e.g.

$$[\text{activation}] = (n_{\text{active}})^4 \tag{6}$$

One result of these assumptions was to provide the experimentally observed sigmoidal activation kinetics and exponential deactivation kinetics.

In Hodgkin and Huxley's original work, they manipulated their measured data so as

to obtain empirical expressions for the individual rate constants $k_1(V)$ and $k_{-1}(V)$. In their reconstruction of the action potential, these expressions were used to calculate the steady-state values of the activation and inactivation parameters, and the effective rate constant $k=(k_1+k_{-1})$. In more recent studies, the analysed data are usually presented in the form of graphs of the voltage dependences of the steady state values and of the time constant tau $=1/k$. Hodgkin and Huxley obtained the best fits to their data by describing I_K with a fourth order activation process and no inactivation, and I_{Na} with a third order activation process and first order inactivation.

(5) An electrical circuit model was then drawn for the squid axon by adding a membrane capacitance, C_m, and a linear "leak" conductance G_l. When the currents are summed at the "inside" node,

$$C_m \frac{dV}{dt} + VG_l + I_{Na} + I_K = 0 \tag{7}$$

which can be rearranged

$$\frac{dV}{dt} = \frac{-1}{C_m}(VG_l + I_{Na} + I_K) \tag{8}$$

for numerical integration. At each time interval the changes in activation and inactivation parameters are calculated and then eqn (8) is used to calculate the change in membrane voltage.

(b) *Action potentials in neurone R15*

For the simulation of the currents flowing during the generation of action potentials in the neurone R15 of *Aplysia californica*, our equations are based on the measurements by Adams and Gage (1979a,b) of I_{Na} and I_{Ca} in neurone R15 of *Aplysia juliana* and those by Thompson (1977) of I_K in *Tritonia*. The parameters in their equations were adjusted in our simulation so as to produce quantitative agreement with our own less extensive measurements of these currents in neurone R15 of *A. californica*.

Action potentials in R15 have rather complex waveforms which, moreover, change during a burst. Thus although they are threshold events, as other action potentials are, it would be inaccurate to describe them as all-or-none events. The rising phases of action potentials in R15 are controlled by influxes first of Na^+ and then later of Ca^{2+}. The conductances for these ions increase with depolarization, leading to a regenerative increase in inward current and a continuing increase in the rate of rise of the action potential. As g_{Na} begins to inactivate, and g_K begins to activate, the rate of rise slows until the action potential reaches its peak amplitude. Thus far the situation is much the same as in the squid axon. In R15, however, a sizeable inward current is carried by Ca^{2+} and the Ca^{2+} current (I_{Ca}) shows only slow inactivation. As a result of this continued inward current flow, the membrane potential falls rather more slowly then in the squid axon. Only when the voltage falls to approximately 0 mV do the Ca^{2+} channels begin to deactivate as a result of their voltage dependence. Deactivation of I_{Ca} results in an increase in net outward current (Fig. 3, right column, I_m), and the resulting more rapid hyperpolarization leaves a distinct "shoulder" on the falling phase of the action potential.

The kinetic description for I_{Na} used by Hodgkin and Huxley in the squid axon also provides a good description for I_{Na} in *Aplysia*. The parameters in our simulation were essentially those of Adams and Gage (1979b), with I_{Na} activating between -20 and $+25$ mV with time constants (reciprocal rate constants) ranging from 5 to 0.5 msec, and inactivating between -30 and $+5$ mV with time constants ranging from 20 to 10 msec. A significant inward current is carried by Ca^{2+} in R15. We chose an exponent of 2 for activation of I_{Ca} (Adams and Gage, 1979b, found best fits to their data with exponents between 1 and 2). The parameter values measured by Adams and Gage (1979b) were adjusted slightly to give better agreement with our measurements in *A. californica*. In our simulation, I_{Ca} activates between -10 and $+35$ mV with time constants ranging from 1 msec to 3 msec and inactivates between -45 and 0 mV with time constants between 150 and 70 msec.

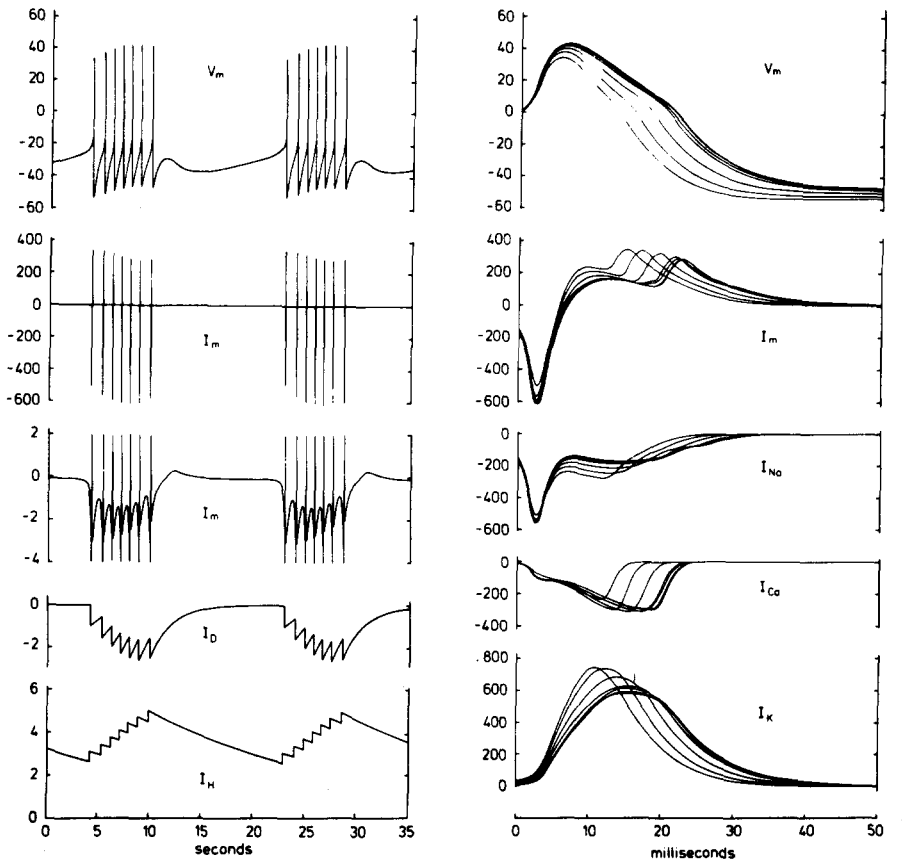


FIG. 3. Membrane potential and current flows during simulation of bursting in R15. Left column: extended time scale to illustrate bursting. Right column: traces were "triggered" at 0 mV to illustrate the current flows during action potential generation. V_m, membrane potential; I_m, total membrane current (on two amplitude scales in left column); I_{Na}, I_{Ca} and I_K, individual ionic currents; I_D, I_H transient depolarizing and hyperpolarizing after-currents. Voltage scales are in mV, current scales in nA.

Among the K^+ currents, I_A appears to be inactivated at the membrane potentials that are produced during bursting (Section III.2) and was omitted from our simulation. Attempts to include I_R are described below. I_K and I_C were combined into a single term and the second order activation kinetics suggested by Thompson (1977) were used. Activation takes place between -15 and $+35$ mV with a time constant that reaches a maximum of 70 msec at -15 mV and decreases at both more positive and more negative voltages.

During a burst, the action potentials change shape in several ways: (1) their duration increases; (2) the undershoot decreases; (3) the positive overshoot increases. The increase in duration appears to be due to inactivation of the K^+ current that mediates repolarization of the action potential (Aldrich *et al.*, 1979). Recovery from inactivation must be quite slow, since action potential durations continue to increase during bursts that are several seconds long. We included a K inactivation term in our simulation to provide inactivation between -30 and $+10$ mV with time constants ranging from 2000 to 10 msec. The results can be seen in Fig. 3 (right column, traces V_m and I_K). Inclusion of inactivation leads to a potassium current (I_K) which is largest for the first action potential in a burst and which activates more slowly and reaches smaller peak amplitudes with later action potentials. As a result, the durations of the action potentials increase during the burst from 14 to 22 msec.

In principle, it is possible that K^+ inactivation could also account for the observed decrease in the action potential undershoot. However, this seems not to be the case. Including very large amounts of K^+ inactivation, by manipulation of its voltage dependence and time constants, caused increases in action potential duration far greater than any recorded in physiological conditions, but produced changes in the undershoot of only 1 to 2 mV. The

reason for this seems to be as follows. At 30–50 msec after initiation of an action potential, when the undershoot reaches its peak, the only significant current flow is carried by K^+ (Fig. 3, right column). Thus, the membrane potential must approach E_K. In fact, changes in E_K have been observed (Eaton, 1972) and attributed to increases in the local concentration of K^+ next to the membrane, perhaps within the numerous infoldings in R15's somal membrane (Section II.1). We have incorporated this phenomenon into our model by assuming that the localized extracellular pool of K^+ increases in proportion to I_K and decreases to approach the bulk $[K^+]_{out}$ of 10 mM according to a simple exponential. We assumed that $[K^+]_{in}$ remained constant at 200 mM (the value measured by Kunze and Brown, 1971) and at each time interval recalculated the K^+ equilibrium potential from the Nernst equation. With a diffusion time constant of 2 sec, the undershoot following action potentials in our simulation resembled closely that seen physiologically (compare Figs 1A and 3, left column, V_m). In the simulation, action potential activity produced changes in E_K from -74 mV during the interburst to -62 mV following the first action potential to -52 mV following the sixth action potential in a burst. Correspondingly, the undershoot decreased from -53 mV after the first action potential to -47 mV after the sixth.

The increase in positive overshoot of the action potentials results in part from K^+ inactivation and changes in E_K, and in part from changes in Na^+ inactivation. The reason for the changes in Na^+ inactivation during the burst is found in its voltage dependence. I_{Na} begins to inactivate in R15 at about -30 mV. Therefore the extent of inactivation depends on the amount of time that the membrane potential remains above this voltage or, conversely, on the rate of depolarization that preceeds the action potential. Prior to the first action potential in a burst, the rate of depolarization is rather small and the largest inactivation results. The maximal inward Na^+ current during the first action potential is about 500 nA (Fig. 3, right column, I_{Na}), as is the maximum net inward current (I_m). For subsequent action potentials, the depolarization preceeding the action potential is more rapid and Na inactivation is less. As a result, I_{Na} is about 50 nA larger. The net inward current, which also reflects the decreases in I_K, grows by about 100 nA, and the positive overshoot increases during the burst from $+34$ to $+43$ mV.

(c) *Bursting in neurone R15*

It is now clear (Section III.2) that cell bodies possess a far greater abundance of conductance pathways than do squid axons. Some of these currents, as discussed in the previous section, account for the complex action potential waveforms that are often observed in cell bodies. Other currents, which may be orders of magnitude smaller than action potential currents, play critical roles in determining the activity of the cell. The reason why such small currents can have such large effects is clear: they are active at membrane potentials where the other currents are not. The interplay of these small currents determines whether, and with what time course, the membrane potential reaches threshold for activation of the larger currents. There seems to be ample teleological justification for the differences between axons and cell bodies. Axons are all-or-none conductors of information from one location to another. Cell bodies are the integrators of information from many sources. The cell body (or cell body-axon hillock) must make a "decision" on whether or not to generate an action potential for subsequent transmission by the axon (Rasmussen, 1976; Treistman, 1980). This "decision" is based on a subtle balancing of many signals including, in the case of R15, signals that are generated endogeneously by the cell body itself. The roles of several of these small currents in generating rhythmic activity in cell R15 are described below.

The major currents controlling bursting are I_D and I_H. I_R and I_{NSR}, which are important in burst modulation (Section IV.3), are incorporated into the model as components of the total background current reflected in the I–V curve. Our initial attempt to include I_R and I_{NSR} as separate currents with time and voltage-dependent parameters revealed a so-far unresolved problem arising from a lack of knowledge about the ways in which these underlying currents change between "steady state" and "bursting state". As discussed previously (Section III.1), the steady state I–V curve can be accounted for by three currents, I_R, I_{NSR}, and a linear leak current (I_L). When used for the simulation of I_R and I_{NSR}, the difference currents found in

blocking experiments (Benson and Levitan, 1983; Barker and Smith, 1978) gave quite good agreement between predicted and observed steady state I–V curves. Moreover, these simulations predicted quite accurately the observed effects of serotonin and dopamine (Section IV.3). A problem arose, however, when an attempt was made to account for the I–V curves that are measured when the cell is in a bursting state. These I–V curves are obtained by allowing the cell to burst freely and then turning on a voltage-clamp to interrupt the burst. If the clamp is left on long enough, as described in Section III.1, the steady-state I–V curve is obtained, with its typical NSR region beginning at -80 to -60 mV. If, however, the measurements of current are made shortly after turning on the clamp (e.g. between 100 msec and 2 sec), the resulting I–V curves are nearly linear. An NSR shows up only at -25 to -20 mV (Adams and Levitan, 1978). We attempted in our simulation to mimic the "bursting state" I–V curve by providing activity-dependent inactivation of I_{NSR} (I_H probably arises from inactivation of at least one component of I_{NSR}—see Section III.2) and by adjusting the relative amplitudes of I_R and the leak conductance. However, we were not able by these means to obtain I–V curves as linear as those observed physiologically, unless I_R and I_{NSR} were reduced essentially to zero. Such a situation would appear to be in contradiction to the effectiveness of serotonin and dopamine in changing the burst patterns by changing I_R and I_{NSR}. It seems clear that resolution of this problem will require experimentation, but the very nature of the problem may make it difficult to devise unambiguous experiments.

In the present version of the simulation, a linear leak conductance has been used to represent all of the "background" currents since the I–V curves for cells in the bursting state are nearly linear. As a corollary, since a voltage-dependent description of the background currents has not been used, voltage-independent representations of the two transient currents, I_D and I_H have also been used. These were modelled as current sources, to which each action potential added a constant increment, and in whieh the summated currents then decayed exponentially to zero. The complete model, incorporating membrane capacitance (C_m), leak conductance (G_l), I_D, I_H and the three action potential currents, I_{Na}, I_{Ca} and I_K is shown in Fig. 4. The current flows I_{Na}, I_{Ca} and I_K were calculated during numerical integration of the circuit equation that describes the model.

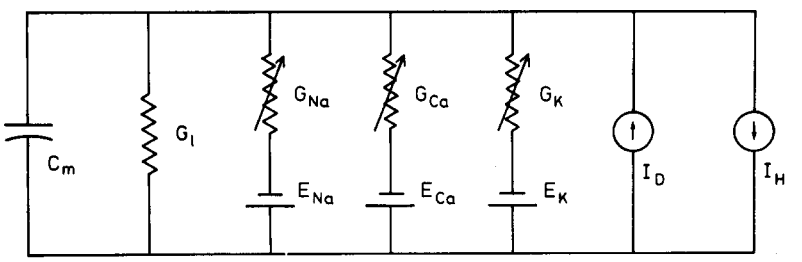


FIG. 4. Electrical circuit model for R15, including membrane capacitance (C_m), leak conductance (G_l), the fast action potential currents, I_{Na}, I_{Ca} and I_K, and the slow transient after currents, I_D and I_H. For the simulation illustrated in Fig. 3, the circuit equation describing this model was integrated numerically (program available upon request). C_m was 50 nF, G_l (10 megohm)$^{-1}$. I_D and I_H were incremented 1 nA and 0.5 nA, respectively by each action potential and decayed with time constants of 2 and 20 sec. I_{Na}, I_{Ca} and I_K were calculated using the voltage dependences and kinetics described in the text.

The predicted activity of the model is shown in the left side of Fig. 3. The membrane potential (V_m) displays many of the characteristics of bursting in R15 (*cf.* Fig. 1A). The total membrane current (I_m) illustrates the changes in action potential currents during the burst that give rise to the observed changes in duration, undershoot and positive overshoot. On an expanded current scale (third trace) the membrane current that underlies generation of the burst pattern can be seen. In this trace, the action potential currents, both inward (I_{Na} and I_{Ca}) and outward (I_K) go offscale by two orders of magnitude (compare second and third traces) and, of these three currents, only the rapidly rising phase of I_{Na} can be seen just preceding each action potential. The currents that are active *between* the action potentials, i.e. the currents

that control the pattern of the burst, are very small compared with the currents that generate the action potentials. For example, the currents during the burst are only -2 to -3 nA, but it is changes in the amplitudes of these currents that give rise to the change in rate of depolarization and action potential frequency during the burst. The rapid "swoop" into the interburst hyperpolarization phase is produced by an outward current of less than 0.5 nA (the positive current at 12 and 31 sec in Fig. 3, left side, third trace). These small subthreshold currents are made up of the current through the linear leak conductance (not shown) and the two transient burst currents I_D and I_H (lower two traces, left side). Between bursts, I_D falls very nearly to zero, the result of a decay time-constant of 2 sec. In this simulation, the summated I_D grew with each of the first five or six action potentials, but then reached a plateau. The summation of I_D increments gives rise to the increase in total inward current between action potentials and the consequent increase in the rate of depolarization and action potential frequency during the burst. In contrast to I_D, I_H does not fall to zero between bursts because its decay time constant (20 sec) is longer than the length of the interburst phase (in this simulation, 13 sec). Instead, it is merely modulated approximately ± 1 nA around a centre value of about 4 nA. The summated value of I_H increases with each action potential in the burst, although not quite linearly, in accord with experimental finding (Adams, 1985). As I_H increases, the net inward current becomes less, and the action potential frequency decreases until finally (after the seventh action potential) the sum of the subthreshold currents ($I_D + I_H + I_L$) is no longer large enough (negative) to depolarize the cell to threshold for another action potential. At this point, the summated I_D is decaying quite rapidly and the "swoop" into the inter-burst hyperpolarization phase is a reflection of the decay of I_D, not of the sudden activation of some outward, hyperpolarizing current. During most of the inter-burst phase, the only currents active in our simulation are I_L and I_H. I_L tends to depolarize the cell and I_H keeps it hyperpolarized. As I_H decays, however, the cell begins to depolarize again, leading into the next burst. In the neurone, it is possible that activation of I_{NSR} contributes to the inter-burst "pacemaker" potential.

The values used for the parameters of I_D, I_H and I_L in this simulation are close to those that are measured experimentally (Adams, 1985). Changes in these values lead to predictable changes in the simulated burst patterns. Increases in the incremental amplitudes of I_D, or in its decay time constant, both lead to an increase in the summated value of I_D during the burst, a prolongation of the burst, and an increase in firing rate during the burst. In addition, since I_H must summate to a larger value in order to terminate the burst, the subsequent interburst recovery phase is prolonged. Increases in the incremental values of I_H lead to shorter bursts and increases in the decay time constant of I_H prolong the interburst phase.

This model of bursting in R15 can be summarized qualitatively as follows. During the generation of action potentials in the R15 soma, both I_{Ca} and I_{Na} contribute to the depolarizing phase of the spike, but I_{Na}, with its faster kinetics, dominates action potential-initiation and rise time (Adams and Gage, 1979a). Repolarization results from the inactivation of I_{Na} and I_{Ca} (Adams and Gage, 1979b), with I_{Ca} deactivation kinetics and voltage-dependence being responsible for the characteristic shoulder on the repolarizing phase. I_K is important in repolarization but it is unclear to what extent I_C contributes since under voltage clamp the depolarizing pulse necessary to admit sufficient Ca^{2+} to evoke I_C is always much longer than an action potential (Thompson, 1977). However, if I_C is strongly voltage-dependent, it may be important in repolarization from the peak depolarization of an action potential (Gorman and Thomas, 1980b). During a burst, action potential broadening is due to inactivation of I_K (Aldrich *et al.*, 1979). This is further evidence of a minor role for I_C which does not inactivate, and, indeed, which should increase as Ca^{2+} accumulates within the cell. The increase in spike frequency is in part due to the increase in I_D (and/or another Ca^{2+}-activated inward current, Hofmeier and Lux, 1981). The decrease in undershoot results from a positive shift in E_K as K^+ accumulates outside the cell with each action potential (Lux and Heyer, 1975; Gorman *et al.*, 1982a). The increase in positive overshoot results from these changes in I_K and E_K but is also due to a decrease in I_{Na} inactivation as the rate of action potential depolarization increases. These changes also contribute to the control of intraburst spike frequency.

The rates of depolarization and repolarization of the action potential are fast in comparison with the inter-burst changes in membrane potential, so that the currents involved in action potential production are hundreds of times larger than those responsible for burst organization and the inter-burst interval.

Examining now these slower changes in membrane potential, we see from Figs 1A and 3 that at the end of the burst the membrane hyperpolarizes and then slowly depolarizes again. The current underlying this hyperpolarization, designated I_H, has been shown previously to have the following properties: (a) it is blocked by Co^{2+} and other calcium channel blockers (Adams *et al.*, 1980a); (b) injection of EGTA, which chelates intracellular Ca^{2+}, reduces or removes this current (Meech, 1974a; Lewis and Wilson, 1982). Since it is known that Ca^{2+} builds up inside the soma during a burst (Thomas and Gorman, 1977), and that Ca^{2+} injection can evoke I_C (Meech, 1972), it has frequently been concluded that activation of I_C underlies the post-burst hyperpolarization (Junge and Stephens, 1973; Johnston, 1980; Carnevale and Wachtel, 1980), and that slow sequestration of intracellular Ca^{2+} results in a slow decline of I_C (Gorman *et al.*, 1982a). Furthermore, the apparent decrease in membrane conductance during the course of the inter-burst interval (Section III.1), has been interpreted as reflecting a slow decline in I_C. For R15, this widely held view is, however, incorrect (Adams, 1984). Experiments under voltage-clamp conditions show that in R15 I_H has the following additional properties: (c) a low concentration of tetraethylamonium (TEA) shown to block I_C leaves I_H unaffected in the same cell, and, indeed, I_H is not blocked by a combination of 50 mM TEA, 5 mM 4-aminopyridine (4AP), and 10 mM Cs^+ (Adams and Levitan, 1985); (d) I_H does not reverse at E_K but remains an outward current (Adams and Levitan, 1985), although it decreases with hyperpolarization in a voltage-dependent manner (Lewis and Wilson, 1982); (e) I_H is unaffected by increases in $[K^+]_0$ (Adams and Levitan, 1985); (f) I_H decreases as $[Ca^{2+}]_0$ is decreased (Adams and Levitan, 1985). From these additional observations, we conclude that the outward current underlying the post-burst hyperpolarization is the I_H as described in Section III.2. I_H arises from a decrease in a calcium current which probably inactivates due to intracellular calcium accumulation (Tillotson, 1979). As calcium is sequestered following the burst, I_H declines, representing a slow increase in a resting calcium conductance (as noted above, this resting current is probably a component of I_{NSR}). Changes in extracellular $[K^+]$ associated with the burst and post-burst hyperpolarization (Gorman *et al.*, 1982a) are most likely due to outward action potential currents and subsequent diffusion.

I_{NSR}, which is partially activated at hyperpolarized potentials, is increasingly activated during the slow depolarization of the inter-burst interval, as a result of its voltage dependence (Gorman *et al.*, 1982a), (and also calcium-sequestration, if I_H and I_{NSR} share the same channels). This creates a positive feedback which produces an increasing rate of depolarization leading to a voltage-dependent inflection and the first action potential.

This first somal action potential initiates an axonal action potential in the axon hillock which propagates away along the axon. The generation of this action potential results in the activation of I_D. As we have discussed in Section III.2, it remains to be determined whether I_D is an ionic current mediated by specific ion channels (perhaps a Ca^{2+}-activated Na^+ current) or is an electrotonic current from the axonal action potential conducted back into the soma. I_D is a depolarizing current of slow kinetics which increases during the burst by summation of the currents activated by each of the first 2 to 5 action potentials (Adams, 1985). The resulting depolarization contributes to the observed increase in action potential frequency during the first half of the burst. I_D then begins to plateau (Fig. 3), and I_H, which increases more slowly, begins to counteract I_D causing the action potential frequency to decrease. Finally, as I_H equals and then becomes larger than I_D, the last depolarization of the burst is the DAP, initiated by I_D and terminated by the influence of I_H combined with a rapid decline in I_D. The DAP is thus not a partially-activated action potential, but a reflection of the current activated by an action potential as it moves away along the axon. The decline of I_D when firing ceases, while I_H is near its maximum causes the post-burst hyperpolarization. I_D declines much more rapidly than I_H (Fig. 2). I_R (not included in the simulation) probably contributes outward

current at the maximum hyperpolarization. It is conceivable that I_A might also contribute to prolongation of this phase of maximum hyperpolarization.

IV. MODULATION OF BURSTING

In the previous section we described how R15 endogenously generates a rhythmic pattern of bursts of action potentials. In the living animal, this activity is modulated by a wide range of inputs. In this section, Part 1 summarizes the known synaptic inputs onto R15, as well as the evidence for the possible effects of light and a circadian clock. Part 2 is an outline of the effects of experimental application of various putative *Aplysia* neurotransmitters, particularly acetylcholine and GABA, together with an outline of the role of neuropeptides in the modulation of bursting activity in R15, and of the effects of other neuroactive compounds. We discuss the evidence that some of these transmitters mediate known synaptic inputs onto R15 and show that they probably act as "classical" neurotransmitters, activating synaptic, non-voltage-dependent currents in the neuropil, remote from the cell body. In Part 3, we deal separately with the effects of serotonin and dopamine, since they act on currents which are part of the burst generating mechanism. In *Aplysia*, they thus differ from the other neurotransmitters which appear to act simply by hyperpolarizing or depolarizing the neurone. The currents affected by serotonin (I_R) and by dopamine (I_{NSR}) therefore play a dual role, taking part in both the burst cycle and its modulation.

1. *Synaptic Inputs, Light and Circadian Rhythmicity*

(a) *Synaptic inputs*

The analysis of the synaptic inputs onto R15 is complicated by the lack of identified presynaptic neurones other than L10. This means that the multitude of synaptic potentials recorded in R15 are for the most part evoked experimentally by stimulation of the various nerve trunks of the abdominal ganglion. Apparently identical PSP's can be evoked by stimulation of different nerves, and stimulation of the same nerve at different intensities and frequencies evokes different PSP's. Furthermore, in most cases several different unitary PSP's are evoked simultaneously and are thus superimposed on one another in the recording from the R15 soma. In this part we will attempt to summarize the known inputs, and also to outline briefly the forms of synaptic plasticity observed at synapses to which R15 is postsynaptic. Finally, we will outline the phase resetting effects of synaptic input and current injection.

Table 2 summarizes the identified inputs onto R15. The primary sources of information for these synapses are Frazier *et al.* (1967), Kandel *et al.* (1967b), and Parnas *et al.* (1974). The amplitudes and time courses of the post-synaptic potentials cover a wide range in magnitude (Chalazonitis, 1968), and some involve conductances purely in the neuropil while others involve modulation of the somal burst currents. Some of the PSP's of neuropilar origin scarcely appear above the electrical noise in a somal recording. Nevertheless, synapses which alter the membrane conductance in the neuropil, while not having a large direct effect on somal activity, may act to augment or diminish responsiveness of the neurone to other synaptic inputs as a consequence of the change in membrane resistance.

Of the identified interneurones, the cell body of only one, L10, has been located. Stimulation of L10 produces an EPSP in R15, the transmitter is most likely acetylcholine, and the post-synaptic conductance activated by the transmitter is a Na current which appears to be a "classical" synaptic current not involved in the burst mechanism. The evidence for the role of acetylcholine at this synapse is given below in Part 2 of this section. So far there is compelling evidence only for acetylcholine to indicate a transmitter function with R15 as the target neurone. It is interesting that while L10 depolarizes R15, the same transmitter released from different branches of L10 hyperpolarizes four bursting cells in the left upper quadrant of the abdominal ganglion (Kandel *et al.*, 1967a).

Interneurone II is a central, multineuronal burst generator which functions as a command centre for respiratory pumping (Byrne and Koester, 1978). Activation of Interneurone II by stimulating the branchial nerve or other abdominal ganglion connectives elicits in R15 an

EPSP followed by a long-lasting hyperpolarization often known as ILD (inhibition of long duration, Tauc, 1958: long-lasting hyperpolarization, LLH, Chalazonitis, 1977; King *et al.*, 1977; Roth *et al.*, 1984). The EPSP is brief and constitutes a minor component of the response. The ILD can last for many minutes. This response appears to be a component of the response to Input III and is discussed in the next paragraph.

Parnas *et al.* (1974) distinguished three inputs onto R15 which were revealed by stimulating the right pleurovisceral connective with successively stronger stimuli (Table 2). The higher threshold inputs normally appeared superimposed on the lower ones, but could be separated pharmacologically and by stimulation via other connectives. The lowest stimulus level activates Input I, which appears to be the same as Interneurone XIII (Frazier *et al.*, 1967), and elicits a characteristically large EPSP, probably cholinergic and exhibiting various forms of synaptic plasticity which have been the subject of extensive study at this synapse (Table 3A and B, and following paragraph). Input I also produces long-lasting excitatory effects on R15 (Parnas *et al.*, 1974) but the mechanism is unknown. Stimulation at a higher intensity (Input II) elicits a biphasic response, a brief EPSP followed by inhibition lasting several seconds. Neither phase of this response is sensitive to d-tubocurarine and the inhibition is almost certainly the result of an increase in Cl^- conductance (Parnas *et al.*, 1974). Stimulation of the right connective at the highest intensity activates Inputs II and III simultaneously. However, Input III appears to activate synaptically Interneurone II, so that the response to Input III can be observed in isolation by stimulating the branchial nerve (Adams *et al.*, 1980b). Input III also produces a biphasic response dominated by ILD which can last for up to 4 hr. The duration of the ILD appears to depend on a mechanism endogenous to R15 and not on prolonged release by the presynaptic cell (Roth *et al.*, 1984). Several hypotheses have been advanced to account for this response in terms of specific ionic currents. Dependence on K^+, Ca^{2+}, cyclic AMP, and temperature, accompanied by a decrease in the negative slope region of the I–V curve are observations common to most reports (Parnas and Strumwasser, 1974; Wilson and Wachtel, 1978; Sawada *et al.*, 1980; Adams *et al.*, 1980b; Treistman, 1980). Our conclusion from the data so far available is that the greater part of the inhibitory response of R15 to stimulation of Input III is due to the activation of I_R(K^+, cyclic AMP) and inactivation of I_{NSR} (Ca^{2+}, temperature, negative slope region) with a contribution from Cl^- as yet not understood. It would be interesting to know whether penicillin affects this response, since penicillin reduces Cl^--mediated responses in *Aplysia* neurones to iontophoretic application of several neurotransmitters (Pellmar and Wilson, 1977). In Part 3 of this section we will discuss the significance of the synaptic modulation of I_R and I_{NSR}, and the question of the identity of the transmitters involved.

Kandel and Tauc (1964, 1965a,b) described heterosynaptic facilitation of the large EPSP recorded in the silent "giant cell" R2 of *Aplysia* in response to stimulation of the right pleurovisceral connective. The similar large EPSP evoked in R15 by stimulation of the same connective (Interneurone XIII; Input I) has proved to exhibit not only heterosynaptic facilitation but also short-term facilitation, frequency facilitation and post-tetanic potentiation (Schlapfer *et al.*, 1974). Since all of these phenomena can be sequentially produced at this synapse, it has been possible to study the properties and interactions of the mechanisms that produce them. The interesting results of these studies are too extensive to be adequately reviewed here but in Table 3A we have attempted to provide a guideline to the publications treating the various forms of plasticity recorded at this synapse, and in Table 3B we have summarized the pharmacology of these processes. The variety of these pharmacological effects is an important reminder that putative neurotransmitters may act not only via the postsynaptic conductances which are the main subject of this review, but also on presynaptic conductances and by modulating the synaptic processes of facilitation, depression and potentiation.

Synaptic input which alters currents localized in the neuropil acts on burst production, which takes place in the soma and proximal axon, in the same manner as current injection into the soma. The soma membrane is depolarized or hyperpolarized and the voltage-dependent burst currents are affected by this change in membrane potential, but they are not modulated directly as in the responses to dopamine or input from Interneurone II. R15 and

TABLE 2. SYNAPTIC INPUTS ONTO NEURONE R15

Presynaptic Element	Postsynaptic Response	Transmitter	Conductance	References
Interneurone I (identified abdominal ganglion neurone L10)	EPSP	Acetylcholine	Na^+	Kandel *et al.* (1967a,b), Frazier *et al.* (1967)
	EPSP accompanied by inhibition			Kehoe (1972)
Interneurone II (activated by stimulation of the branchial nerve—but similar effects from other connectives)	EPSP followed by inhibition of long duration (ILD)	ILD—not dopamine	I_{NSR}	Frazier *et al.* (1967), Tauc (1968), Stinnakre and Tauc (1969), Waziri and Kandel (1969), Jacklet *et al.* (1970), Roth *et al.* (1984)
Right pleurovisceral connective including the axon of Interneurone XIII. Similar effects from left connective stimulation	Input I—large EPSP			Frazier *et al.* (1967), Gerschenfeld *et al.* (1967), Audesirk (1973), Koester *et al.* (1974), Segal and Koester (1980), Parnas *et al.* (1974), Bryant *et al.* (1973)
	Input II—EPSP followed by ILD			Parnas *et al.* (1974), Adams *et al.* (1980b)
	Input III—same as Interneurone II	ILD—not dopamine	I_{NSR}	Parnas *et al.* (1974), Adams *et al.* (1980b), Parnas and Strumwasser (1974)

See also Table 3.

TABLE 3. (A) PLASTICITY IN THE RIGHT CONNECTIVE-R15 LARGE EPSP

Measured Variables	Modulatory Mechanism	References
Depression and frequency facilitation	Regulation of transmitter availability	Schlapfer *et al.* (1974)
	Presynaptic control of transmitter release	Woodson *et al.* (1975)
Frequency facilitation and post-tetanic potentiation	Interaction of increased transmitter availability and release with pool depletion	Schlapfer *et al.* (1976), Woodson *et al.* (1976a)
Depression	Four parameter control of transmitter availability	Woodson *et al.* (1976b)
Depression, frequency facilitation, and post-tetanic potentiation	Heterosynaptic inhibition	Woodson *et al.* (1976d)
	Presynaptic stimulus frequency and duration	Grenon and Tremblay (1983)
Post-tetanic potentiation	Heterosynaptic input	Newlin *et al.* (1980b)
	Rhythmic presynaptic activity	Woodson and Schlapfer (1979)
	Temperature	Schlapfer *et al.* (1975)

(B) PHARMACOLOGY OF PLASTICITY IN THE RIGHT CONNECTIVE-R15 LARGE EPSP

Pharmacon	Effects	References
Protein synthesis inhibitors	None	Schwartz *et al.* (1971)
Serotonin (low concentration)	Increased rate of decay of post-tetanic potentiation	Newlin *et al.* (1980a)
Serotonin (high concentration)	Reduced EPSP amplitude; reduced depression	Tremblay *et al.* (1976)
Dopamine	Increased rate of decay of post-tetanic potentiation; reduced EPSP amplitude; reduced depression	Newlin *et al.* (1980a), Tremblay *et al.* (1976)
Ca^{2+} and Mg^{2+}	High Ca^{2+} increases facilitation	Halstead and Jacklet (1974)
Memantine	High concentrations block transmission	Klee (1982)
Ethanol	Accelerates the rate of decay of post-tetanic potentiation	Traynor *et al.* (1976), Woodson *et al.* (1976c)
Morphine	Decreased EPSP amplitude	Tremblay *et al.* (1976), Waziri (1976)
Benzodiazepines	Decreased EPSP amplitude, depression and post-tetanic potentiation; increased frequency facilitation	Tremblay and Grenon (1980)
Pentobarbital	Reversibly decreases the EPSP amplitude	Barker and Gainer (1973)
GABA	Decreased EPSP amplitude, depression and post-tetanic potentiation; increased frequency facilitation	Tremblay and Plourde (1977), Tremblay and Grenon (1980), Philippe *et al.* (1981)

TABLE 3. (B) continued

Pharmacon	Effects	References
Hemicholinium-3	Decreased EPSP amplitude; actions on post-tetanic potentiation and depression depend on $[Ca^{2+}]$	Tremblay *et al.* (1980), Stinnakre (1970)
Baclofen	Reduced EPSP amplitude; increased frequency facilitation and reduced depression	Philippe *et al.* (1981), Tremblay and Philippe (1980)
Cholinergic agents	Effects indicated four classes: 1. Agents with no effect 2. Curare-like post-synaptic blockers 3. Carbachol-like depressors of early EPSP's (reduce transmitter release) 4. Trimethidinium-like depressors of later EPSP's (reduce transmitter supply to readily releaseable pool)	Woodson *et al.* (1975)

other molluscan bursting neurones respond to this kind of stimulus almost as physical relaxation oscillators, phase-resetting and often showing transient changes in burst period. Strumwasser (1965, 1967a) first drew attention to the phase-shifting effects of current injection into R15. Mathieu and Roberge (1973) analysed the relation between burst duration and burst period. They found only a weak correlation between the two parameters, but Pinsker (1977a) showed that experimentally changing the number of action potentials in a burst (i.e. burst duration) altered the length of the following inter-burst interval and hence the burst period. In Section II, we showed how action potential production is central to the generation of the burst cycle. Bryant *et al.* (1973) found that the R15 burst cycle could be entrained by periodic synaptic input, and Kim and Kim (1977) investigated entrainment of R15 by sinusoidal current injection. Chalazonitis *et al.* (1982) describe the differences in afterdischarge response to electroshock among R15, tonically active cells, and silent cells. Utilizing *Aplysia* bursters very similar to R15 but with inhibitory input from L10, Pinsker carried out an extensive analysis of these phase-resetting and entrainment properties in response to periodic synaptic input (Pinsker, 1977a,b,c; Pinsker and Kandel, 1977; Pinsker and Bell, 1981). The result is an interesting and useful application of the concepts and terminology of physical oscillator theory to a neuronal oscillator, as well as a further confirmation of the importance of action potential production in the burst cycle. However, caution is necessary when attempting to apply these observations to the confirmation of particular hypotheses of the ionic basis of bursting. Many neuronal bursters exhibit the behaviour of relaxation oscillators (e.g. crustacean cardiac bursting neurones, Benson, 1980a) while having quite different ionic burst generators (Benson and Cooke, 1984).

(b) *Light response*

In common with extra-ocular neurones in many molluscs, some of the neurones in the abdominal ganglion of *Aplysia* are light-sensitive (Arvanitaki and Chalazonitis, 1961a). The giant neurone R2 responds to light by hyperpolarizing as a result of activation of a K^+ current (Brown and Brown, 1972, 1973). The response requires the release of calcium from lipochondria or pigmented granules present in the cytoplasm (Henkart, 1975; Brown *et al.*, 1975). The ventral photoresponsive neurone (v.p.n.) in the same ganglion is a thousand-fold more sensitive to light than R2 but has a similar peak to its action spectrum (470 nm) (Andresen and Brown, 1979; Andresen *et al.*, 1979). The v.p.n. response is also K^+-mediated, the photo-current having neither time nor voltage dependence. The K^+ photo-current is non-rectifying (Andresen and Brown, 1979). R15 is also light-sensitive but the response is less

thoroughly studied than in R2 or v.p.n. (Arvanitaki and Chalazonitis, 1961b; Frazier *et al.*, 1967). R15 hyperpolarizes when illuminated and membrane conductance increases. The peak of spectral sensitivity is 485 nm but it is yet not determined whether or not a K^+ current mediates the response (Nelson *et al.*, 1976). Light appears to have some long-term effect on the burst frequency of R15. This is discussed in the next part in relation to changes in activity over hours and days *in vitro*.

(c) *Circadian rhythmicity*

Strumwasser (1963, 1965) noticed that the burst frequency of R15 varied considerably during the course of hours-long intracellular recordings, and he attributed this variation to control of the burst frequency by a circadian clock in R15 (Strumwasser, 1963, 1965, 1967a,b, 1968, 1971, 1973). Further experiments seemed to confirm this conclusion, although the considerable variation in the time of occurrence of peak frequencies often had to be attributed to "seasonal modulation" (Lickey, 1969, 1971; Lickey *et al.*, 1971; Audesirk, 1973; Lickey and Zack, 1973; Audesirk and Strumwasser, 1975). Nelson *et al.* (1976) reported a circadian rhythm in the amplitude of the light response described above. Most of these experiments were limited to little more than a single day of recording and in many cases temperature variation was $\pm 1°C$ or more. Several cycles of "free-running" rhythmicity in constant light and temperature constitute the conventional test for true endogenous clock control, and the extreme temperature sensitivity of the burst mechanism (I_{NSR} in particular) has been described in Section III.2. Long duration recording from R15 has failed to provide evidence for circadian rhythmicity. Beiswanger and Jacklet (1975) made continuous intracellular recordings of longer than a week, and Lickey *et al.* (1976) recorded the activity of R15 extracellularly for up to 40 days. Neither laboratory could find evidence for a circadian rhythm of action potential or burst frequency. There does, however, seem to be a relationship between the time of the initial peak of activity, the time of dissection, and the preceding light–dark cycle, but the significance of this relationship is unclear (Beiswanger and Jacklet, 1975; Audesirk and Strumwasser, 1975). Burst frequency in R15 can be driven rhythmically by synaptic input (Beiswanger and Jacklet, 1975; Woodson and Schlapfer, 1979) but this rhythmicity does not persist when the rhythmic input ceases (Beiswanger and Jacklet, 1975). Possible origins of circadian rhythmic input *in vivo* are reviewed by Lickey *et al.* (1976). The general conclusion to be drawn from the work reviewed here is that in the isolated abdominal ganglion, bursting in R15 is not under circadian clock control but that burst frequency is easily influenced by light and temperature. In the intact animal, there are inputs capable of modulating the burst frequency in R15 and some of these could be under circadian control or direct control of light, salinity, and other environmental parameters (Strumwasser *et al.*, 1972).

2. *Neurotransmitters, Neuropeptides and Other Neuroactive Compounds*

In the previous section (IV.1), the known synaptic inputs onto R15 were outlined. Here we consider the actions of various transmitter substances and the ionic basis of their effects, where this is known. We also consider the evidence available on whether a particular compound might be a transmitter mediating the synaptic effects described above, and whether it modulates bursting in the manner of a "classical" transmitter, by hyperpolarizing or depolarizing the neurone via non-voltage-dependent channels in the synaptic region, or whether it alters the burst currents themselves. Acetylcholine is the only compound for which there is strong evidence for a physiological function in synaptic input onto R15, and among the peptides, the effect of egg-laying hormone is most likely of physiological significance. The mode of action on R15 of serotonin and dopamine have been intensively studied, and their effects on somal currents will be described in the next part.

(a) *Acetylcholine*

Tauc and Gerschenfeld (1961, 1962) tested a large number of *Aplysia* neurones for responses to acetylcholine and found that some cells depolarized (D-cells) and others

hyperpolarized (H-cells). Acetylcholine always depolarizes R15 or, in the case of a voltage-clamped R15, evokes depolarizing current. The response is rapid to peak and of brief duration (Blankenship *et al.*, 1971), and is blocked by *d*-tubocurarine (Kandel *et al.*, 1967b; Carpenter *et al.*, 1977; Woodson *et al.*, 1975). It is also blocked or reduced by Tris (Wilson *et al.*, 1977), pentylenetetrazol (Park and Klee, 1975), and morphine (Waziri, 1976; *cf.* Tremblay *et al.*, 1974). Acetylcholinesterase inhibitors have no potentiating effect on the response of R15 to experimentally applied acetylcholine (Fossier *et al.*, 1983).

It is very likely that acetylcholine is the neurotransmitter released by L10 onto R15 to evoke the EPSP. The evidence can be summarized as follows. L10 has the enzymes necessary for acetylcholine production, and indeed does manufacture acetylcholine (McCaman and Dewhurst, 1970; Giller and Schwartz, 1971a; Eisenstadt *et al.*, 1973; McCaman *et al.*, 1973; Koike *et al.*, 1974). As discussed above, L10 synapses directly onto R15 producing an EPSP when stimulated. This EPSP is blocked by *d*-tubocurarine (Gerschenfeld *et al.*, 1967; Kandel *et al.*, 1967b; Kehoe, 1972; Woodson *et al.*, 1975; Parnas *et al.*, 1974). The EPSP is also blocked by hexamethonium (Kehoe, 1972; Schlapfer *et al.*, 1974), a compound which blocks Na^+-dependent acetylcholine responses (Tauc and Gerschenfeld, 1962). R15 contains acetylcholinesterase which suggests that it is cholinoceptive (Giller and Schwartz, 1971b). The acetylcholine response and the EPSP have similar reversal potentials (usually 0 to -20 mV) (Blankenship *et al.*, 1971), and both are mediated predominantly by a change in Na^+ conductance. Cl^- and K^+ are not involved (Blankenship *et al.*, 1971; Carpenter *et al.*, 1978). Taken together these results strongly suggest that the EPSP evoked in R15 by L10 has acetylcholine as its neurotransmitter. It remains to be shown that L10 actually releases acetylcholine onto R15 when it is stimulated.

The acetylcholine receptors on the soma and in the neuropil produce an identical response to iontophoresed acetylcholine. It seems likely, however, that the physiologically functional receptors for acetylcholine are those in the neuropil region where the synapses occur (see Section II.1), because the Na^+ current evoked by acetylcholine appears to be a "classical" non-voltage-dependent, synaptic current (Blankenship *et al.*, 1971). As far as we can determine, none of the somal burst currents described in Section III is sensitive to acetylcholine. This suggestion receives support from the observation that iontophoretic application of acetylcholine to the soma produces a depolarization whose effect is no greater than of a similar transient depolarization produced by experimental current injection. Acetylcholine thus has a mode of action on R15 similar to that of the synaptic inputs onto non-bursting neurones.

(b) *Gamma-amino butyric acid (GABA)*

GABA induces a slow depolarizing response in R15 (Tremblay and Plourde, 1977). It reaches a peak in about 9 sec and lasts for an average of 1.5 min (Yarowsky and Carpenter, 1977, 1978; Carpenter *et al.*, 1978). This response can be elicited in the neuropil only. There seem to be no GABA receptors on the cell body and the evoked current appears to be remote from the soma (Yarowsky and Carpenter, 1978). The response to GABA consists of a conductance decrease in those cases where it was large enough to be measured and the evoked current is hypothesized to be carried by K^+ ions, since neither Na^+ nor Cl^- is involved, but Yarowsky and Carpenter (1978) were unable to reverse the response at the K^+ equilibrium potential. The response decreases with hyperpolarization. Although the effect of iontophoresed GABA can last for up to 2 min, the response of R15 is not long term. As with acetylcholine, GABA appears to act via a purely synaptic current. However, to our knowledge, no systematic investigation has been carried out intc the possibility of GABA-induced changes in the known burst currents.

(c) *Other compounds*

Our knowledge of the responses of R15 to glutamate, aspartate, octopamine and noradrenaline is very limited. Glutamate produces a hyperpolarization which is rapid to peak and of brief duration (Carpenter *et al.*, 1978). This response depends on extracellular Cl^- concentration, involves an increase in membrane conductance and has a reversal

potential near the estimated Cl^- equilibrium potential (Carpenter *et al.*, 1978). The glutamate response is facilitated by aspartate which has no effect by itself (Carpenter *et al.*, 1978).

Noradrenaline inhibits bursting in R15 (Boisson and Chalazonitis, 1972) possibly via dopamine receptors (see Section IV.3). Octopamine and phenylethanolamine, possibly acting on a single receptor type, hyperpolarize R15 (Carpenter *et al.*, 1978) but the mechanism of the response is unknown.

Molluscan neurones have been extensively tested for their responses to convulsants and anticonvulsants. Trimethadione at high concentrations decreases bursting by increasing the interburst hyperpolarization, and can eventually silence the cell (Kreisman *et al.*, 1978). The effect of trimethadione is thought to be of presynaptic origin rather than a direct action on R15. Pentobarbital at 200 μM does not alter the I–V curve of R15 over the limited range of potentials studied (Barker and Gainer, 1973). High concentrations of furosemide, a diuretic, alter the Ca^{2+}-dependent component of the somal action potential in R15 (Section III.2) and also alter the Cl^--dependent responses of other *Aplysia* neurones to GABA and acetylcholine (Scappaticci *et al.*, 1981). The convulsant pentylenetetrazole blocks the R15 acetylcholine response at millimolar concentrations (Park and Klee, 1975; Park, 1976), and decreases I_C (Ewald and Eckert, 1983). At 35 mM, it hyperpolarizes R15 (Klee, 1976). Strychnine, another convulsant, reduces total outward current during depolarizing steps in voltage-clamped R15 (Klee *et al.*, 1978) and alters the burst pattern in the unclamped neurone (Klee, 1976).

High concentrations of ethanol (0.4 to 0.8 M) abolish bursting activity and the NSR of the I–V curve (Section III.1) in R15 (Silver and Treistman, 1982; Schwartz, 1983).

(d) *Neuropeptides*

Peptides can markedly alter the activity of neurones through their effects on voltage-independent ("classical synapse") and voltage-dependent (e.g. burst cycle) currents. The action of peptides on various molluscan neurones has been reviewed by Barker (1976) and Barker and Smith (1977). Here we will confine our attention to the two known classes of peptide effect on R15. The first is the augmentation of bursting in R15 by egg laying hormone (ELH) from the bag cells, and the second includes the effects of the vertebrate hormones oxytocin and vasopressin.

The bag cells are neuroendocrine cells occurring as a cluster at the base of each of the two anterior connectives of the abdominal ganglion (Frazier *et al.*, 1967; Coggeshall, 1967). When experimentally stimulated by a brief train of electrical pulses, they synchronously discharge with a prolonged burst of action potentials which lasts about 20 min (Mayeri *et al.*, 1979b). Kupfermann (1970) showed that, among others, a 6 K dalton peptide, now known as ELH, is released during this bag cell discharge. The bag cells appear to release their contents into the vascular spaces of the abdominal ganglion (Kupfermann, 1970) and by this means the ELH reaches other abdominal neurones including R15. The evidence for this hypothesis is that a brief (1 min) application of partially purified ELH has a prolonged (up to 3 hr), burst-augmenting action on R15, increasing the number and frequency of spikes per burst and the amplitude of the interburst hyperpolarization (Branton *et al.*, 1978a), which appears to be identical to the effect on R15 of the bag cell discharge (Branton *et al.*, 1978b; Mayeri *et al.*, 1979b; Maat *et al.*, 1983). There is no structural evidence for a synaptic route between the bag cells and R15, and the slow, smooth onset is typical of a hormonal effect rather than a classical fast synaptic mechanism. ELH affects other neurones in different ways (Mayeri *et al.*, 1979a) but in all cases the effect of bag cell discharge or ELH application lasts many times longer than the stimulus. These effects differ from the excitation mediated by classical synapses not only in their duration. Maintained depolarization or hyperpolarization of R15 or other target neurones does not mimic the response. It seems likely that ELH acts on voltage-sensitive currents intrinsic to the burst cycle (Mayeri, 1979), but no voltage-clamp data are currently available. The response appears to be localized in or near the soma in neurones L3 and L6 (Brownell and Mayeri, 1979) where ELH-induced prolonged inhibition is mediated by an increase in g_K and by a slower ionic process that is relatively independent of

membrane conductance (Brownell and Mayeri, 1979). Although the function of ELH as a promoter of egg-laying in *Aplysia* is known, the physiological significance of its action on R15 is still a matter for speculation.

Barker *et al.* (1975) found that lysine-vasopressin, oxytocin and arginine-vasotocin, as well as certain analogues, have apparently selective effects on bursting activity in cell 11 in *Otala*, the homologue of R15. At 10^{-9} to 10^{-8} M, the peptides increased the amplitude and duration of the interburst hyperpolarization and, as with ELH, these effects long out-lasted the period of application. Based on observations made under voltage-clamp (Barker and Gainer, 1974), it is possible that these compounds act by increasing I_{NSR}. The significance of these responses is unknown but the low concentration required to elicit the effects suggests that they are physiological. In *Aplysia*, none of the three peptides affected the neurones R2, R14, L2, L6, L7 or L11, but all three were reported to increase bursting activity in R15 (Barker *et al.*, 1975). Moore *et al.* (1981) have also shown that arginine-vasotocin augments the bursting in R15 and they have been able to locate an arginine-vasotocin-like peptide in the anterior ganglia of *Aplysia*. Luteinizing hormone releasing hormone (LHRH) and thyrotropin-releasing hormone (TRH) are without effect on R15, although LHRH has an excitatory effect on other "white cells" of the *Aplysia* abdominal ganglion (Seaman *et al.*, 1980).

3. *Mode of Action of Serotonin and Dopamine*

The effects of serotonin and dopamine on R15 have been intensively studied, and we will summarize the accumulated observations and conclusions from this work. It must be kept in mind that these compounds have completely different actions on other molluscan neurones, even within the same *Aplysia* ganglion. It has yet to be shown whether either serotonin or dopamine is released as a transmitter or neurohormone with R15 as a target. We will describe how the inhibitory synaptic input onto R15 evoked by stimulation of Interneurone II (Section IV.2 and Table 2) is a mixed synapse modulating I_{NSR} and I_R, the two currents affected by dopamine and serotonin respectively (Fig. 5), and we will discuss the small amount of evidence available concerning whether either of these compounds is involved as a transmitter at this synapse.

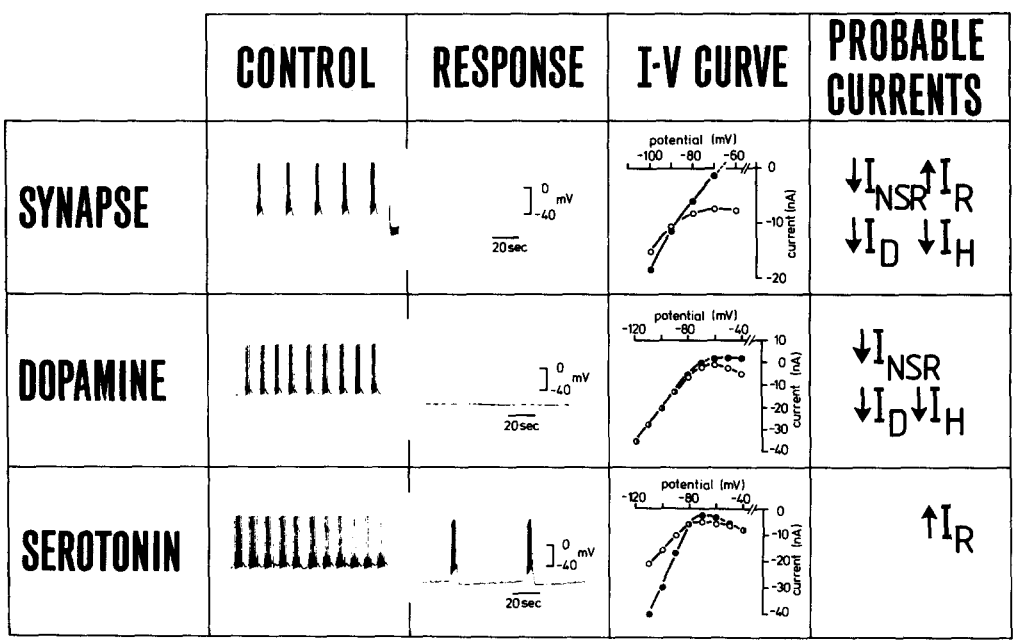


FIG. 5. Changes in neural activity and in steady-state *I*-*V* curves that result from synaptic stimulation, or from addition of dopamine or serotonin to the bathing medium.

(a) *Serotonin*

The application of serotonin (5-hydroxytryptamine) to R15 results in a hyperpolarization of slow onset and long duration (Drummond *et al.*, 1980a). At 10^{-5} M, serotonin can inhibit bursting completely. Transient excitatory effects produced by very high serotonin concentrations may be non-physiological (*cf.* Stefani and Gerschenfeld, 1969; Tremblay *et al.*, 1976; Kandel, 1976). The hyperpolarizing response of R15 to serotonin is mediated by cyclic AMP (Drummond *et al.*, 1980a; Benson *et al.*, 1980) and protein phosphorylation is an essential step in the molecular transduction between the serotonin receptor and the K^+ channel whose activation results in hyperpolarization of the neuronal membrane (Adams and Levitan, 1980, 1982b). R15 is a neuronal system in which all the necessary criteria (Robison *et al.*, 1971; Greengard, 1976) have been fulfilled to assign definitively to cyclic AMP the role of "second messenger". In this section we will review briefly the evidence that cyclic AMP is the second messenger involved in the R15 serotonin-response, and then we will describe the ionic mechanisms by which serotonin alters the balance of somal burst currents in R15.

The biochemical, pharmacological and electrophysiological experiments summarized here were almost all carried out specifically on R15. For example, some biochemical experiments involved dissecting out the cell body of R15 while others required injection of radio-labelled compounds into R15. This experimental specificity is essential since molluscan neurones display a number of responses to serotonin, some involving K^+ currents (Deterre *et al.*, 1981, 1982; Klein and Kandel, 1980) and others not (Gerschenfeld and Paupardin-Tritsch, 1974; Pellmar and Carpenter, 1981). Cyclic AMP or cyclic AMP-dependent protein phosphorylation have been implicated in at least four other responses in *Aplysia* (Klein and Kandel, 1980; Siegelbaum *et al.*, 1982; Pellmar, 1981a,b; Pellmar and Carpenter, 1981; Eskin *et al.*, 1982; Benson, 1980b; Drake and Treistman, 1980b, 1981; Kaczmarek *et al.*, 1978) and three in *Helix* neurones (Deterre *et al.*, 1981, 1982; De Peyer *et al.*, 1982; Swandulla and Lux, 1984). These systems have been reviewed elsewhere (Kehoe and Marty, 1980; Kennedy, 1983; Levitan *et al.*, 1983b; Drummond 1983). Here we have confined our attention to R15.

Evidence for cyclic AMP as a second messenger in the R15 response to serotonin is as follows:

(1) Serotonin hyperpolarizes R15. It does so by activating I_R, the anomalously rectifying K^+ current (Benson and Levitan, 1982, 1983, 1985). Serotonin does not increase I_A in R15 (Benson and Levitan, 1983), and as far as is known, I_R is not activated by any other transmitter. However, it should be noted that there is evidence that both serotonin and cyclic AMP increase I_C when the latter is evoked in R15 by injection of Ca^{2+} (Ewald and Eckert, 1983) and injection of cyclic AMP activates a sodium current in R15 and other molluscan neurones (Connor and Hockberger, 1984).

(2) Serotonin activates an adenylate cyclase attached to the R15 soma membrane (Levitan, 1978c) and this cyclase is linked to a receptor specific for serotonin. Pharmacological analysis of isolated-membrane serotonin receptors, utilizing whole abdominal ganglia rather than solely R15, shows that the serotonin receptor mediating the hyperpolarization of R15 is indistinguishable from the serotonin receptor associated with the adenylate cyclase (Drummond *et al.*, 1980b; Levitan and Drummond, 1980). This means that the same serotonin receptor mediates hyperpolarization and increases intracellular cyclic AMP. Inactivation of the adenylate cyclase by injection of GDPβS blocks the action of serotonin (Lemos and Levitan, 1984).

(3) Increased intracellular cyclic AMP hyperpolarizes R15 and does so by activating I_R, the same current that serotonin activates. The hyperpolarization is obtained no matter which of the following means is used to increase intracellular cyclic AMP:

(a) Extracellular application or intracellular injection of permeant, phosphodiesterase-resistant, cyclic AMP analogues (Treistman and Levitan, 1976a; Levitan and Norman, 1980; Drummond *et al.*, 1980a; Treistman, 1981).
(b) Application of phosphodiesterase inhibitors. These compounds prevented or reduced the break-down of cyclic AMP thereby mimicking or enhancing the effect of serotonin

(Treistman and Levitan, 1976a; Drake and Treistman, 1980b; Levitan and Norman, 1980; Drummond *et al.*, 1980a).

(c) Intraneuronal injection of guanylyl-imidodiphosphate, an activator of *Aplysia* adenylate cyclase (Treistman and Levitan, 1976b; Treistman and Drake, 1979; Treistman, 1981; Lemos and Levitan, 1984).

(4) It has been proposed that cyclic AMP acts exclusively by activation of cyclic AMP-dependent protein kinases (Kuo and Greengard, 1969). Protein kinase inhibitor (PKI) is a naturally-occurring, 10 kD protein (Walsh and Ashby, 1973) which binds with high affinity to the active catalytic subunit of cyclic AMP-dependent protein kinase and inhibits its phosphorylating activity in various species (Walsh and Ashby, 1973) including *Aplysia* (Adams and Levitan, 1982b). When PKI was injected into R15 it completely inhibited the serotonin response, but not the response to dopamine (Adams and Levitan, 1980, 1982b). Injection of other proteins into R15 had no effect on the serotonin response (Adams and Levitan, 1982b). These observations demonstrate that the blockade of the serotonin response is produced specifically by PKI and that PKI-induced blockade is selective for the serotonin-induced increase in I_R.

When all of these lines of evidence are taken together, there is little room for doubt that the hyperpolarization of R15 produced by serotonin is mediated by cyclic AMP and most probably by cyclic AMP-dependent protein phosphorylation. Serotonin causes the phosphorylation of a number of specific proteins in R15 (Lemos *et al.*, 1982). It appears that there are a considerable number of substrates for cyclic AMP-dependent protein kinases, as well as for Ca^{2+}- and cyclic GMP-dependent protein kinases, and not all of these are associated with the response to serotonin. There is evidence for an interaction between Ca^{2+} and cyclic nucleotide-induced alterations in I_R (Drake and Treistman, 1980a). Dopamine and synaptic input (branchial nerve stimulation) also induce protein phosphorylation (Lemos *et al.*, 1982, 1984; Novak-Hofer *et al.*, 1985). Work is in progress to determine whether any of these proteins are associated with the I_R K^+-channel (Lemos *et al.*, 1985b). Various aspects of the work outlined here have been reviewed in more detail elsewhere (Levitan, 1978a,b; Levitan *et al.*, 1979, 1983a,b; Levitan and Drummond, 1980; Levitan and Benson, 1981; Levitan and Adams, 1981; Benson *et al.*, 1983; Lemos *et al.*, 1985a).

Now we will consider the ionic mechanism of action of serotonin. It has been found that serotonin activates a K^+ current (Drummond *et al.*, 1980a) and this current has been identified as I_R, the anomalously rectifying K^+ current (Benson and Levitan, 1982, 1983, 1985). As discussed in Section II.2, I_R is partially activated in the non-serotonin treated neurone. It is an imperfect rectifier which means that there is a small outward contribution from I_R to the net membrane current (the I–V curve) at membrane potentials (V_m) between 5 and 35 mV more positive than the potassium equilibrium potential (E_K). This is important because in physiological conditions the V_m is never negative to E_K, so that the small outward current is the part of I_R on which serotonin exerts its effect. As we have emphasized in Section III.1, at V_m between -80 mV and -50 mV, the "steady-state" I–V curve passes close to the zero current axis, on the inward current side (Fig. 1B). Since these V_m's are subthreshold for the majority of the large, voltage dependent currents, the current active in this V_m range will not affect the action potentials themselves, but will control the pattern of their generation during bursting and especially during the inter-burst interval. Figure 5 shows how a small increase in the outward I_R can bring the I–V curve closer and ultimately up to the zero current axis. At low serotonin concentrations, activation of I_R produces a net decrease in total (inward) membrane current over the V_m range traversed by V_m during the inter-burst interval of the burst cycle, causing the inter-burst hyperpolarization to be greater. At a high serotonin concentration, the net membrane current can be reduced to zero by the increase in outward I_R. The neurone is thereby rendered stably silent at the hyperpolarized V_m at which the I–V curve crosses the zero current axis with a positive slope (usually at V_m -75 mV to -65 mV). The activation of I_R by serotonin is different from the response designated "B-response", an increase in g_{K+} by some *Aplysia* abdominal ganglion neurones, since the latter appears not to involve rectification (Gerschenfeld and Paupardin-Tritsch, 1974).

Figure 5 summarizes the evidence that an increase in I_R is a component in certain synaptically induced current changes in R15. This raises the question of whether serotonin is the synaptic transmitter involved in increasing I_R. In Section IV.1, we described the long lasting inhibition induced in R15 by stimulation of various abdominal ganglion connectives (Table 2). Parnas and Strumwasser (1974) and Adams *et al.* (1980b) showed that in the branchial nerve there are apparently at least two pre-synaptic elements with similar stimulus thresholds which produce inhibition of long duration in R15. From the I–V curves in Fig. 5, it appears that the current changes due to high intensity, repetitive stimulation of the branchial nerve can be broken down into an increase in I_R and a decrease in I_{NSR}. The responses of I_R to serotonin and to stimulation of Input III (Interneurone II) appear to be the same, and in the presence of cyclic AMP, stimulation of Input III gives only a decrease in I_{NSR}. The phosphosdiesterase inhibitor theophylline enhances synaptically-induced hyperpolarization in R15 (Treistman and Levitan, 1976b) and IBMX potentiates the hyperpolarizing phase but does not increase the duration of inhibition (Roth *et al.*, 1984). Is serotonin the transmitter for this input? At the moment no further data are available on which to base a decision. No serotonergic neurones are known to be presynaptic to R15, and R15 is not distinguished by a superior ability to break down serotonin (Goldman and Schwartz, 1977). Identity of serotonin and the synaptic transmitter cannot be assumed, even though both have the same effect of increasing I_R. However, since the receptor–adenylate cyclase complex described here was isolated from the soma of R15 which is also the locus of the I_R channels, it is possibly significant that many somata in the abdominal ganglion are apposed by serotonergic fibres bearing varicosities and possibly indicating release of serotonin directly onto the somal membrane (Kistler *et al.*, 1983; Ono and McCaman, 1984).

(b) *Dopamine*

The effects of dopamine on R15 are well-known. Ascher *et al.* (1967) saw depolarizations and hyperpolarizations in response to dopamine in various *Aplysia* neurones, but Ascher (1968, 1972) showed that R15 always hyperpolarizes, often for many minutes, when exposed to dopamine. Local application of dopamine is effective on the proximal axon but not on the soma (Ascher, 1972). Figure 5 shows that when dopamine is applied to a voltage-clamped R15, dopamine has the effect of reducing I_{NSR} (Boisson and Gola, 1976, 1978; Wilson and Wachtel, 1978; Gospe and Wilson, 1980). Lewis and Wilson (1983), and Lewis *et al.* (1984) provide evidence that dopamine reduces Ca^{2+} influx (presumably by reducing I_{NSR}) in the proximal axon but not in the soma. It can be seen from Fig. 5 that the result of this decrease in I_{NSR} is to reduce the magnitude of the negative slope region (NSR; Section III.1 and III.2) of the I–V curve, so that it passes through the zero current axis, with a positive slope, at a membrane potential which is below threshold for the voltage-dependent action currents. As with serotonin, this means that the membrane potential will tend to stabilize at the potential at which the I–V curve crosses the axis. This is observed experimentally (Fig. 5), and the neurone remains silent (e.g. Boisson and Chalazonitis, 1976) until the effect of the dopamine decreases to the extent that I_{NSR} is sufficiently activated to add the inward current necessary to move the I–V curve below the zero current axis. The neurone is thus unstable again and can re-enter its burst cycle. As with serotonin, the ionic mechanism of action of dopamine depends on a balance of subthreshold currents producing a net current near zero. We have suggested in Section III.3 that I_{NSR} is not simply a "synaptic" current, but is voltage-dependent and plays an important role in burst generation. Dopamine thus acts not just by hyperpolarizing the neurone as a classical IPSP but by modulating one of the more important burst currents. Dopamine also reduces I_H, another important burst current, concomitantly with its reduction of I_{NSR} (Adams *et al.*, 1978; W. B. Adams, unpublished observations; Lewis *et al.*, 1984). This finding is consistent with our hypothesis that I_{NSR} and I_H are carried by the same channels. I_D appears also to be affected by dopamine. If I_D does not have its own channels, this observation would have to reflect a change in the electrotonic spread of current to the soma. Low concentrations (1–5 μM) of dopamine enhance the total outward current following a burst of spikes (Anderson *et al.*, 1984). Chesnoy-Marchais (1984) reports furthermore that 1 min exposures to 100 μM dopamine increase a resting K^+

current in R15. This current remains to be investigated in detail. However, for the long duration response it is most likely by its action on I_{NSR}, a current activated in the hyperpolarized range of membrane potentials, that dopamine can silence R15.

As we have described earlier in this section, reduction of I_{NSR} is one component of the ionic basis of the long-lasting inhibition of R15 resulting from stimulation of particular synaptic inputs (Fig. 5). Wilson and Wachtel (1978), working with the abdominal ganglion burster L6 under voltage-clamp, showed in some detail that reduction of I_{NSR} can underlie inhibition of long duration. Similar observations have been made for R15 (Adams *et al.*, 1980b; Boisson and Gola, 1976). It appears that the synaptic- and dopamine-induced reductions in I_{NSR}, recorded in the soma, result from conductance changes at an axonal site.

Since both dopamine and a synaptic input have their effects by reducing I_{NSR}, an obvious question is whether dopamine is the synaptic transmitter (Chalazonitis, 1978). The answer appears to be negative, on pharmacological grounds. The pharmacology of the dopamine receptors has been investigated and they have been found to differ markedly from other dopamine and adrenergic receptors (Gospe and Wilson, 1981; Gospe, 1983). Agonists of potency similar to dopamine include norepinephrine, epinephrine, phenylephrine and various analogues of dopamine. Methysergide, ergotamine and other ergot alkaloids irreversibly block the action of dopamine and the other agonists mentioned above (Gospe and Wilson, 1981). Tyramine, β-phenethylamine and *d*-amphetamine hyperpolarize R15 by the same ionic mechanism as dopamine, possibly by presynaptically inducing transmitter release (Gospe and Wilson, 1982). However, compounds such as *d*-LSD and dihydroergotamine, which block the effects of dopamine and the dopamine-like agonists, do not block the inhibitory synaptic input which, as we have shown, affects I_{NSR}, the same ionic conductance as dopamine (Gospe and Wilson, 1982; Adams *et al.*, 1978; Ascher, 1972). The interpretation of these results is complicated by the observation that *d*-LSD and dihydroergotamine are also partial serotonin agonists when applied to R15 (Drummond *et al.*, 1980a). Nevertheless, these findings suggest that a single ionophore can be coupled with two distinct receptor types, and that dopamine might not be the neurotransmitter mediating Input III (Interneurone II).

In summary, we have shown in Section IV.3 how serotonin, dopamine and at least two synaptic inputs produce in R15 long lasting inhibitory responses which persist for many minutes after the termination of the input stimulus. They all have a similar effect on bursting activity, namely to deepen and lengthen the interburst interval, and, at sufficient intensity, to silence the cell. The ionic bases of these responses differ in their details. Dopamine, serotonin and the high threshold Input III (and Interneurone II) decrease I_{NSR} and increase I_R, yet both processes depend for their efficacy on the same principle. Both depend upon the ability of a small net outward current, activated in the subthreshold range where the I–V curve passes close to the zero current axis and where the large magnitude action potential and burst currents are not active, to produce a proportionately large change in net membrane current and ultimately a net membrane current of zero. Furthermore, I_{NSR} is involved not only in modulation by dopamine and synaptic input, but is also an integral part of the burst generating mechanism. It seems to be a characteristic of R15 that burst currents, including the voltage-dependent I_{NSR}, are the effectors of synaptic input and of the action of putative neurotransmitters.

V. SUMMARY

We have concentrated our review on the single identified neurone R15 from the abdominal ganglion of *Aplysia*. With the exception of the squid giant axon, R15 is probably the most widely studied cell in neurobiology. It has long attracted attention because of its bursting activity, its production of peptides, and its possible role in osmoregulation. More recently, a number of studies have focused on the synaptic inputs to R15 and on the mediation by second messengers of its response to neurotransmitters.

A central feature of the functioning of R15 is its rhythmic activity. This activity is generated endogenously by R15, but is subject to modulation by synaptic input and by hormonal

influences. Our account of the burst cycle emerges from a consideration of currents experimentally measurable in R15. Three elements not emphasized in previous descriptions of neuronal bursting activity are very important in our interpretation: (a) individual currents appear to have dual functions taking part both in the burst cycle and in its modulation by external input; (b) very small currents (nanoampere range) can have profound effects on activity in a cell producing action currents that are measured in microamperes; (c) axonal action potential conduction and hence cell topology play a crucial role in the generation of somal activity. These properties reveal a delicate balance of channel thresholds, passive membrane properties and anatomy. This is not an engineer's solution to the problem of endogenous burst production, but a surprisingly complicated harmony of the diverse elements available. This became increasingly clear as we examined some of the modulatory mechanisms present in R15. Many cases are now known of synaptic or neurotransmitter modulation of voltage-dependent currents which contribute to endogenous neuronal activity (Siegelbaum and Tsien, 1983), and R15 provides one of the most complex examples (Wilson, 1980). In R15 the known synaptic inputs, as well as two putative neurotransmitters, directly alter the currents which we have shown are intimately involved in the generation of burst activity.

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